Process Biotechnology

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BIOPROCESS TECHNOLOGY FOR THE PRODUCTION OF CELL BIOMASS AND PRIMARY/SECONDARY METABOLITES

MANUFACTURE OF BAKER'S YEAST

In human diets, microbial protein constitutes a relatively minor portion of the total protein consumed. This primarily comes from edible macrofungi (mushrooms, truffles, etc.) and a small contribution in the form of yeast in bread, estimated at 2g protein per person per week in the western European diet.

A major fermentation industry has developed to manufacture the vast quantity of baker's yeast required for making bread and associated bakery products. The 'skimming method' was one of the first methods employed for the commercial production of baking strains of *Saccharomyces cerevisiae*. This method used media derived from cereal grains, and was similar to brewing and distilling fermentation processes. Here the yeast floated to the top of the fermentation and was skimmed off, washed and press-dried. However, during World War I, due to shortages of cereal grains, the yeast industry sought alternate fermentation materials.

In Germany a process was devised whereby molasses, ammonia and ammonium salt were used in place of cereal-based media. These remains, with certain refinements and automation, as the basis of current methods of manufacture that now generate 2×10^{9} kg/annum worldwide.

Baker's yeast production commences with propagation of a starter culture, which originates from a pure freeze-dried sample or agar-medium culture. Yeast cells are initially transferred to small liquid culture flasks, then on to larger intermediate vessels before being finally used to inoculate the large production fermenters of $50m^3$ to $350m^3$ capacity. Overall, this may involve up to eight scale-up stages to produce the necessary final inoculum volume.

Medium for the production fermentation normally contains molasses as the carbon and energy source, which may be pretreated with acid to remove sulphides and heated to precipitate proteins. Molasses is often deficient in certain amino acids, and supplements of biotin and pantothenic acid are usually necessary. Further nitrogen sources (ammonium salts or urea) may be added, along with orthophosphate and other mineral ions, and the pH is adjusted to 4- 6.

The main objective of the process is to generate a high yield of biomass that exhibits an optimal balance of properties, including a high fermenting activity and good storage properties. Aerobic fermentation favours a high biomass yield, as approximately 50% of the available carbon can be potentially converted to biomass. The maximum theoretical growths yield (Ys) is 0.54 g/g, whereas under anaerobic conditions this value is reduced to around 0.12 g/g. Therefore, the aim is to maintain the cells under aerobic conditions. This is partially achieved by strong aeration of

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the fermentation broth, which is usually further increased as the fermentation progresses, via an aeration system that must achieve an oxygen transfer rate of 150 mmol/L/h. In addition, the fermentation is operated as a fed-batch process. The nutrients are added at a specific rate to prevent the operation of the Crabtree effect, a phenomenon where substrate suppresses aerobic respiration. This feeding regime limits anaerobic metabolism and ethanol production, which would otherwise result in lower biomass yields. By maintaining a specific growth rate (m) of 0.2–0.25/h at 28–30°C, cell concentrations of up to 60 g/L can be achieved. At the end of the growth phase, when all the nutrients are depleted, the aeration is continued for a further 30min to 'ripen' the yeast. This encourages production of the storage carbohydrate trehalose, reduces protein and RNA synthesis, and stabilizes the cells to give a longer storage life.

Centrifugal separators, run at a minimum speed of 5000rpm, are used to remove the yeast from the fermentation broth. Harvested cells are then washed several times with water, chilled to 2–4°C and finally dried to around 70–75%(w/w) moisture using vacuum filter dehydrators. The yeast is usually packed in 1kg blocks and kept under refrigeration. Alternatively, the yeast may be dried further to 7–10% (w/w) moisture to form dried yeast, which can then be stored for long periods without refrigeration. Overall, the cycle of operations, from the initial pure yeast sample to the yeast blocks ready for sale, takes about 2 weeks, passing through 60 generations. Desirable features of baker's yeast and areas where improvements are being sought via genetic engineering include:

- High glycolytic activity, a key feature being CO_2 generation rates or 'gassing power' ,
- Rapid utilization of maltose, which is the main sugar of bread dough,
- Osmotolerance, i.e. the ability to function in the presence of high levels of sugars and salt within dough,
- Good storage characteristics, including cryotolerance(the suitability for frozen storage) and
- High growth rates.

INDUSTRIAL ETHANOL

Most regions of the world have traditionally produced alcoholic beverages from locally available substrates. Similar alcoholic fermentations are now used in some countries to produce fuel grade or chemical feedstock ethanol. The annual world production of ethanol is over 30 billion litres, approximately 70% of which is produced by fermentation, the remaining being mostly manufactured by the catalytic hydration of ethylene. Almost 12% of the fermentation ethanol is beverage alcohol, 20% is for various industrial uses and the remaining 68% is fuel ethanol. Ethanol is an attractive fuel because it may be used alone or mixed with other liquid fuels, e.g. 'gasohol', a blend of 10–22% (v/v) ethanol with gasoline. In 1970s, Brazil and few other countries undertook the full-scale production of ethanol from indigenous renewable biomass resources to offset the growing costs of oil imports. The ethanol was produced by fermentation of sucrose, derived from sugarcane, using *Saccharomyces cerevisiae.* Brazil is now responsible

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for over 46% of annual world production, some 14.5 billion litres of ethanol. However, this is not sufficient to keep up with the growing demand for fuel. Failure to develop their production processes has resulted in the need to import ethanol from the USA and other producing countries.

Apart from sucrose, other conventional fermentation substrates for ethanol fermentations include simple sugars derived from plants and dairy wastes. These require relatively little processing (Fig.1).

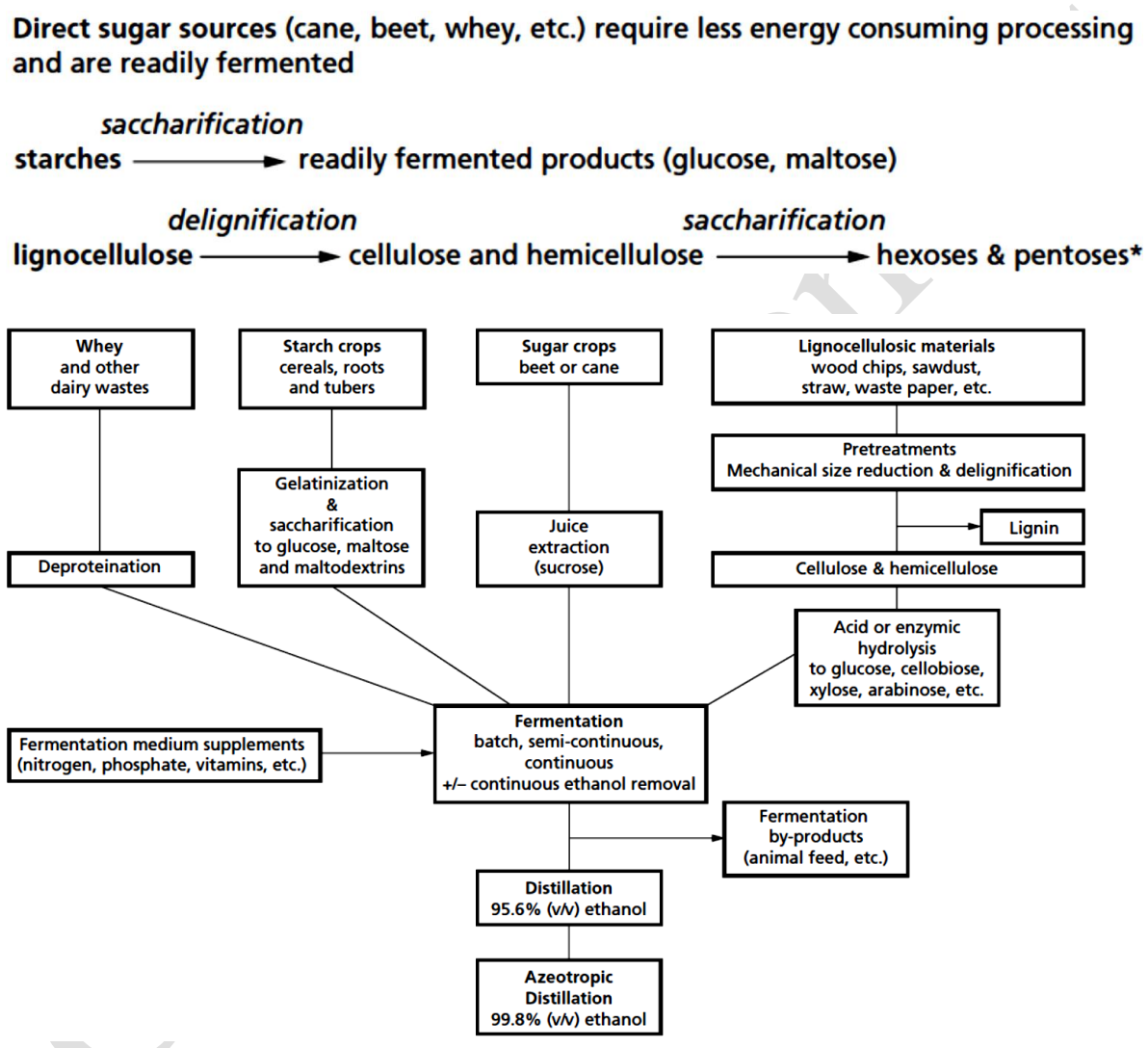


Fig.1 Industrial ethanol production from various substrates.

However, use of root and tuber starch (cassava, potato, etc.) or grain starch (maize, wheat, rice, etc.) demands energy-consuming processing operations to achieve hydrolysis. Even greater processing is necessary prior to the utilization of ligno-cellulosic plant materials.

BIOCONVERSION OF MAIZE STARCH

In North America, wet or dry milling processes have been developed for maize processing to separate corn oil from the starch. This also generates byproducts that can be used for animal feed. Extracted starch is subjected to gelatinization and saccharification, and the resultant sugars can then undergo alcoholic fermentation. The technology initially used was largely based on previously developed for alcoholic beverage production, but now the processes are much more efficient. Enzymatic saccharification is used for converting starch into fermentable sugars employing various thermostable amylases, including gluco-amylases. Fermentation of the resulting sugars, mostly glucose, is carried out by selected strains of *S. cerevisiae* at 32–38°C and pH 4.5–5.0. Fermentations may be batch or continuous process, often with some form of cell recycle, which reduces both the fermentation time and the amount of substrate 'wasted' in conversion to unwanted biomass. Operations under vacuum, which facilitate continuous removal of ethanol to reduce ethanol inhibition and even cell immobilization have been tried.

These alcoholic fermentations generate a 'beer' containing approximately 10% (v/v) ethanol from which the yeast is usually separated prior to distillation. Recovered ethanol may then be dehydrated. The cost of this ethanol recovery is often up to 50% of the total process expenditure. Process byproducts include methanol, glycerol and higher alcohols, such as amyl, butyl and propyl alcohols. Possible alternate approaches to ethanol recovery include the use of continuous extractive fermentation processes using non-volatile, non-toxic solvents, such as oleyl alcohol, which have a high affinity for ethanol. This strategy is useful in overcoming end-product inhibition. The solvents employed are continuously introduced into the fermenter and rise through the medium to form a layer that is continuously removed. Passing through a centrifuge results in the separation of ethanol laden solvent from media and cells, which are then returned to the fermenter. The ethanol may be recovered by flash vaporization and the non-volatile solvent is reused.

Although *S. cerevisiae* is still extensively used for the alcoholic fermentation of simple sugar substrates, there are other organisms with commercial potential. These include bacterial genus *Zymomonas*, such as *Z. mobilis*, which are Gram-negative facultative anaerobes that normally ferment only glucose, fructose or sucrose. They afford greater ethanol yields than does *S. cerevisiae*, but are not ethanol tolerant. In future, alternate routes are likely to involve genetically engineered organisms that have the ability to utilize a wider range of carbon sources and have better fermentation properties. For example, *Escherichia coli*, which normally produce only relatively small amounts of ethanol, have been transformed with a plasmid that encodes the

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alcohol dehydrogenase and pyruvate decarboxylase from *Z. mobilis*. Such transformants produce ethanol under both aerobic and anaerobic conditions.

BIOCONVERSION OF LIGNOCELLULOSIC MATERIALS

Unlike energy crops (cereals, sugar cane and beet, etc.), lignocellulosic plant wastes (sawdust, wood chips, straw, bagasse, waste paper, etc.) have no direct food use. They are renewable resources yet to be fully exploited. Billions of tonnes of these cellulosic materials currently go to waste each year, which could be converted into chemical energy or other useful fermentation products. Lignocellulose is composed of the following polymers.

- **Lignin** (10–35%, w/w), a polymer of three phenolic alcohols (*p*-coumaryl, sinapyl and coniferyl alcohols) that encrusts the cellulose. This material cannot be degraded by microorganisms under anaerobic conditions, but may be used as sources of vanillin, catechol, dimethylsulphide (DMS) and dimethyl sulphoxide (DMSO) when degraded via chemical processes.
- **Cellulose** (15–55%, w/w), a linear homopolymer of β-1,4-linked glucose units. Once hydrolysed, the resultant glucose is readily fermented by many microorganisms, but few can directly utilize the native polymer.
- **Hemicellulose** (25–85%, w/w), a class of heteropolymers containing various hexoses (d-glucose, d-galactose and d-mannose) and pentoses (l-arabinose and d-xylose). Xylose is the second most abundant sugar in nature after d-glucose and may constitute up to 25% of the dry weight of some woody trees, but only a few microorganisms ferment pentoses to ethanol. Importantly, ethanol production from lignocelluloses is economically viable only if both pentoses and hexoses are fermented.

Few microorganisms can utilize lignocellulose directly and those that do, such as some species of *Clostridium*, produce little or no ethanol. Therefore, direct microbial fermentation of cellulosics to ethanol is a distant opportunity. An approach more likely to prove successful in the shorter term involves several steps. First, pretreatment of the lignocellulosic material is necessary prior to the saccharification of hemicellulose and cellulose components. Sugars resulting from chemical and/or enzymatic hydrolysis may then be fermented to produce ethanol, which can be separated from the aqueous phase by distillation.

Pretreatment and saccharification must be conducted in a way that maximizes subsequent bioconversion yields and minimizes the formation of potentially inhibitory compounds, especially furfurals and soluble phenolics. Most lignocellulosic materials require a pretreatment to render the cellulose and hemicellulose more amenable to acid or enzyme hydrolysis. Pretreatment requirements vary with the feedstock and are often substantially less for processed materials such as paper and card. Methods employed include mechanical size reduction by milling, chemical pulping, acid hydrolysis, alkali treatment, auto-hydrolysis, solvent extraction,

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steaming and steam explosion (explosive decompression following high-pressure steam treatment at 4000kPa for 5–10 min).

Various combinations of pretreatment processes may be used depending upon the source of lignocellulosic materials. Some achieve partial saccharification, but further treatment with acid or enzymes is usually necessary. Acid hydrolysis is generally carried out with dilute acid (e.g. 0.5–5% (v/v) sulphuric acid) under pressure to achieve elevated temperatures of 100– 240°C. This treatment is relatively inexpensive, but also generates large quantities of degradation byproducts and undesirable inhibitory compounds. Strong acid hydrolysis often uses concentrated hydrochloric acid at ambient temperature, which gives the highest sugar yield of any acid hydrolysis process. However, such operations are highly corrosive and almost complete acid recovery is essential to make the process economically viable. Acid hydrolysis of mixtures of cellulose and hemicellulose is difficult to control. Hemicellulose is more readily hydrolysed than cellulose and generates sugars early in the process. These sugars may undergo further breakdown to inhibitory compounds, e.g. furfurals. Consequently, conditioning of hydrolysates may be necessary in order to remove these compounds prior to fermentation.

The sugars generated by hydrolysis are primarily glucose, cellobiose (a disaccharide composed of β-1,4-linked glucose units) and xylose. Fermentation of xylose is problematic. *S. cerevisiae*, which is currently responsible for 95% of all ethanol produced by fermentation, does not ferment this monosaccharide. Those organisms that do are not ethanol tolerant and give poor ethanol yields.

There are a number of possible ways by which *S. cerevisiae* could be employed in the alcoholic fermentation of xylose.

- Isomerization of the aldo-sugar, D-xylose, to the keto form D-xylulose, which *S. cerevisiae* can ferment. This may be achieved by carrying out the yeast fermentation in the presence of a bacterial xylose isomerase.
- Genetic engineering of *S. cerevisiae*, to express genes for either: a bacterial xylose isomerase, e.g. from species of *Actinoplanes*, *Bacillus*, etc.: or xylose reductase and xylitol dehydrogenase from a pentose fermenting yeast, e.g. species of *Candida*, *Pichia*, etc. However, there are likely problems with cofactor imbalances with this option. *Z. mobilis* has also been genetically engineered to ferment xylose and may play a future role in the production of ethanol from plant biomass.

AMINO ACIDS

Several amino acids are produced in commercial quantities via direct fermentation processes using overproducing microbial strains or by microbial biotransformation. They are mostly employed as food or animal feed supplements and flavour compounds. However, several amino acids also have uses in pharmaceuticals and cosmetics, and in the chemical industry for the manufacture of polymers.

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L-GLUTAMIC ACID

Of all the amino acid production processes, that of L-glutamic acid is probably the most important in terms of quantity. Its main use is as the flavour enhancer, monosodium l-glutamate (MSG), which can heighten and intensify the flavour of foods without adding significant flavour of its own. MSG is naturally present in certain foods and was discovered to be the 'active' component of a traditional flavour-enhancing seaweed stock used in Far Eastern foods. This compound was first isolated from the seaweed, *Laminaria japonica*, in 1908. Commercial production in Japan followed almost immediately, using extracts of soya protein and wheat gluten. In 1959 the US Food and Drug Administration (FDA) classified MSG as 'generally regarded as safe' (GRAS) due to its history of safe use, and the Joint Food and Agriculture Organization (FAO) ,World Health Organization (WHO) Expert Committee on Food Additives (1970) gave the acceptable daily intake as 0–120 mg/kg body weight. Since the early 1960s, the classical production methods using plant sources have largely been replaced by fermentation processes, which are now responsible for an annual production in excess of 400 000 tonnes. The price of MSG in international trade is an average of US$1.20/kg and apart from extensive use in oriental foods; it is added to a wide range of food products, particularly soups, gravies, sauces and snacks. Glutamic acid-producing microorganisms include species of the closely related genera *Arthrobacter*, *Brevibacterium*, *Corynebacterium*, *Microbacterium* and *Micrococcus*. These are Gram-positive, biotin requiring, non-motile bacteria that have intense glutamate dehydrogenase activity.

Species of

$$\text{Oxoglutaric acid} + NH_4^+ + NAD(P)H + H^+ \xrightarrow{\textit{Glutamatedehydrogenase}} \text{L-glutamic acid} + NAD(P)^+ + H_2O$$

Brevibacterium and *Corynebacteriu* are used for most industrial fermentations. The wild-type *Corynebacterium glutamicum*, for example, exhibits feedback inhibition when cellular glutamic acid concentrations rise to 5% on a dry weight basis. However, the production strains developed using mutagenesis and selection programmes are both regulatory and auxotrophic mutants. These strains have been developed with a high steady-state cytoplasmic amino acid concentration. They accumulate about 30% l-glutamic acid and yield 1 mol of glutamate per 1.4 mol of glucose and importantly are phage resistant. More recently, recombinant DNA technology has been used to increase the activity of specific biosynthetic enzymes by transformation with multicopy plasmids bearing the structural genes for these enzymes. The overall strategy for achieving overproduction of the amino acid involves:

- Increasing the activity of anabolic enzymes;
- Manipulation in regulation to remove feedback control mechanisms;
- Blocking pathways that lead to unwanted byproducts;
- Blocking pathways that result in degradation of the target product; and

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- Limiting the ability to process the immediate precursor of l-glutamic acid, namely oxoglutaric acid, to the next intermediate of the tricarboxylic acid (TCA) cycle, succinyl coenzyme A (CoA), i.e. use of mutants lacking oxoglutaric acid dehydrogenase. During the growth phase, these mutants produce essential intermediates from isocitrate via the glyoxylate cycle.

In addition, as these bacteria do not normally secrete glutamate, ranges of treatments are employed to render the cells more permeable and aid release of the amino acid into the medium. These treatments include: biotin limitation, restriction of phospholipid biosynthesis by adding C_{16}–C_{18} saturated fatty acids during the growth phase, and inclusion of surfactants (e.g. Tween 40) and penicillin in the production media.

INDUSTRIAL PRODUCTION OF L-GLUTAMIC ACID

Industrial-scale fermenters are normally stainless steel stirred tank reactors of up to 450m^3. These are batch processes, operated aerobically at 30–37°C, the specific temperature depending on the microorganisms used.

Apart from carbon and nitrogen sources, the fermentation medium normally contains inorganic salts, providing magnesium, manganese, phosphate and potassium, and limiting levels of biotin. *Corynebacteria* are nutritionally fastidious and may also require other vitamins, amino acids, purines and pyrimidines. The preferred carbon sources are carbohydrates, preferably glucose or sucrose. Cane or beet molasses can be used, but the medium requires further modification as their biotin levels tend to be too high. This can be overcome by the addition of saturated fatty acids, penicillin or surfactants which promote excretion. The nitrogen source (ammonium salts, urea or ammonia) is fed slowly to prevent inhibition of l-glutamate production. Medium pH is maintained around 7–8 by the addition of alkali, otherwise the pH progressively falls as the l-glutamate is excreted into the medium. Accumulation of l-glutamic acid does not become apparent until midway through the fermentation, which normally lasts for 35–40 h and achieves l-glutamic acid levels in the broth of 80 g/L.

Product recovery involves separation of the cells from the culture medium. The l-glutamic acid is then crystallized from the spent medium by lowering the pH to its isoelectric point of pH 3.2 using hydrochloric acid. Crystals of l-glutamic acid are then filtered off and washed. MSG is prepared by adding a solution of sodium hydroxide to the crystalline l-glutamic acid followed by recrystallization.

L-LYSINE

L-Lysine is not synthesized by humans and other mammals. This ‘essential’ amino acid must be taken as part of their diet. However, many cereals and vegetables are relatively low in lysine. Consequently, food products and animal feeds derived from these sources are often

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supplemented with this amino acid. The annual world production of L-lysine necessary to fulfill these requirements is now in excess of 380 000 tonnes. Over 90000 tonnes of this lysine are currently produced by direct microbial fermentation and biotransformation methods. The remaining portion is produced by chemical synthesis. However, this route has the major disadvantage that a mixture of the d- and l-isomers is synthesized, but it is only l-lysine that the body utilizes. Thus, optical resolution is required following chemical synthesis, whereas microbial production has the advantage that only the L isomer is formed.

INDUSTRIAL PRODUCTION OF L-LYSINE

Metabolic control of l-lysine production in wild-type *C glutamicum* is shown in Fig. 2.

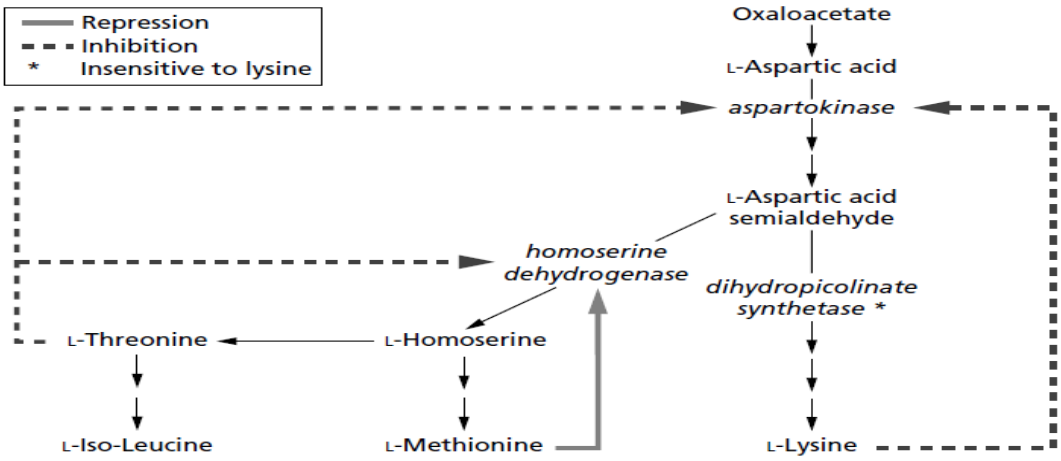


Fig.2

The first key step of this metabolic pathway, aspartate to aspartyl phosphate, catalysed by aspartokinase, is controlled via feedback inhibition by two end-products of this branched pathway, lysine and threonine. Homoserine dehydrogenase activity is also subject to feedback inhibition by threonine and repression by methionine. However, dihydropicolinate synthetase is not inhibited by lysine accumulation, which is unusual for the first enzymes following the branch point of a pathway. The over-producing strains of *C. glutamicum* selected for lysine production have defects in these feedback control mechanisms. They lack homoserine dehydrogenase activity and are thus homoserine auxotrophs. These auxotrophs convert all aspartate semialdehyde to lysine, and because of the lack of threonine synthesis, there is no longer feedback control. However, carefully measured amounts of threonine, methionine and isoleucine must be added to the culture medium to enable this auxotrophic bacterium to grow.

Most commercial l-lysine fermentations are operated as batch processes in aerated stirred tank reactors. Cane molasses is the preferred carbon source, although other carbohydrates, acetic acid or ethanol can be used, often supplemented with soya bean hydrolysates. The temperature is held at 28°C and the pH is maintained at, or near, neutrality by feeding ammonia or urea, which also

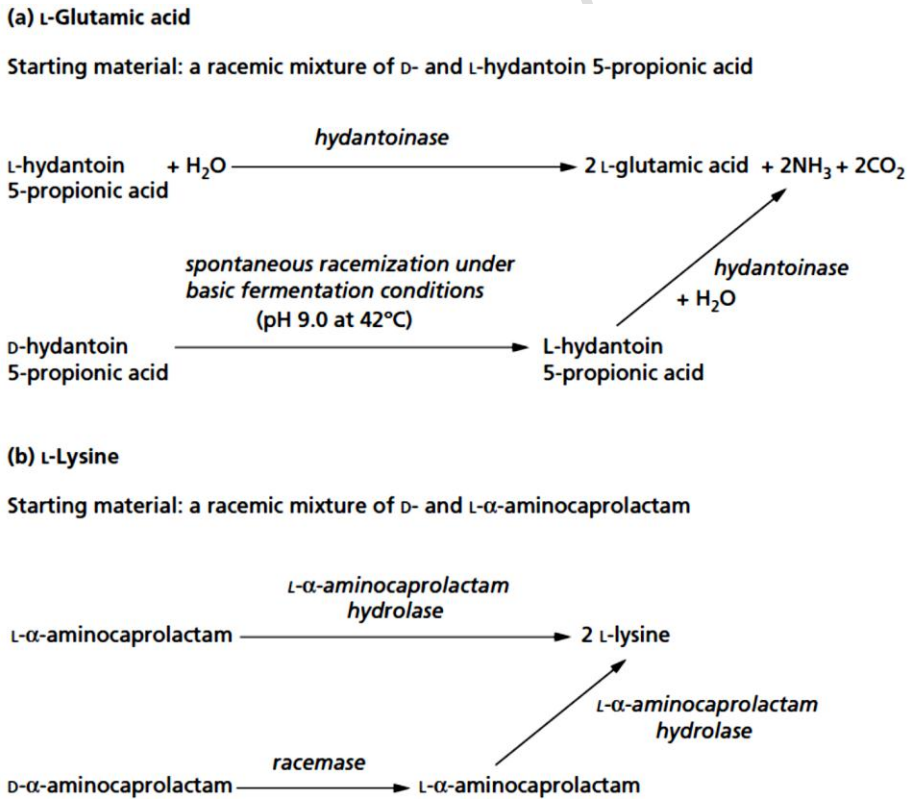
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act as a nitrogen source. Control of the biotin level is very important, as concentrations below 30mg/L result in the accumulation of l-glutamate instead of l-lysine (see l-glutamic acid production above). However, cane molasses usually contains sufficient biotin to fulfill this requirement.

The lag phase is shortened by using a high concentration of inoculum, normally about 10% (v/v) of the fermentation volume. Production of lysine starts in the early exponential phase and continues through to the stationary phase. These fermentations last about 60h and yield 40–45 g/L l-lysine from a molasses concentration of 200 g/L, containing 100 g/L sucrose. Lysine recovery is relatively simple. Once the cells have been removed, the fermentation medium is acidified to pH 2.0 with hydrochloric acid and the l-lysine is adsorbed onto a cation-exchange column in the ammonium form. A dilute solution of ammonia is then used to elute l-lysine from the column. This eluate is reacidified and the product is finally crystallized as l-lysine hydrochloride.

ALTERNATE BIOTRANSFORMATION METHODS FOR THE PRODUCTION OF AMINO ACIDS

Production of l-amino acids may also be performed by the enantio selective hydrolysis of racemic precursors. L-Glutamic acid, for example, can be produced from chemically synthesized d-l-hydantoin 5-propionic acid. This process uses *Bacillus brevis*, which produces the necessary hydantoinase and gives 90% yield. (Fig. 3a).



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l-Lysine can also be produced via a batch biotransformation process from d-l-α-aminocaprolactam, an inexpensive starting material chemically derived from cyclohexane. d-l-α-Aminocaprolactam is added to a reaction vessel at a concentration of 100 g/L, along with the yeast *Cryptococcus laurentii* and the bacterium *Achromobacter obae.* The result is an almost complete conversion of the substrate to l-lysine. This method exploits the stereo specificity of a hydrolase found in *C. laurenti* to convert the l-isomers of the substrate to l-lysine. The remaining d-isomer of α-aminocaprolactam is brought back into the production pathway by a racemase in *A. obae* (Fig. 3b). Similar methods are also available for the synthesis of d-amino acids, some of which are important side-chain precursors for semisynthetic penicillins and cephalosporins, e.g. d-*p*hydroxyphenylglycine.

Fig.3a-3b: Production of l-amino acids by biotransformation.

CITRIC ACID

Citric acid is widely used in the food industry as an acidulant and flavouring agent in beverages, confectionery and other foods. As a food constituent, its use is unrestricted because it has GRAS status. This organic acid also has many non-food applications. They include roles in maintaining metals in solution for electroplating, as a cleaning and 'pickling' agent for metals, and as a replacement for polyphosphates in the detergent industry, along with several pharmaceutical uses.

Until the 1920s citric acid was mainly prepared from lemon juice, but in 1923 Pfizer began operating a fermentation-based process in USA. The production organism was the filamentous fungus *Aspergillus niger*, an obligate aerobe, which was grown in surface culture on a medium of sucrose and mineral salts. Virtually all the worldwide output is now produced by fermentation, which is primarily located in Western Europe and China. Citric acid has become one of the world's major fermentation products, with an annual production of over 550 000 tonnes and a value approaching US$800 million. The demand for citric acid is still increasing, particularly for beverage applications. Surface methods are still operated, but since the late 1940s, submerged fermentations have become the principal mode of production. Many microorganisms, including filamentous fungi, yeasts and bacteria, can be used to produce this primary metabolite. However, *A. niger* still remains as the predominant industrial producer. Specific strains have been developed for various types of fermentation processes, which are capable of generating high yields of citric acid, often in excess of 70% of the theoretical yield from the carbon source.

CITRIC ACID BIOSYNTHESIS

The metabolic pathways involved in citric acid biosynthesis are the Embden–Meyerhof–Parnas (EMP) pathway and the TCA cycle. *A. niger* also operates the pentose phosphate pathway, which can compete with glycolysis for carbon units.

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The first stages of citric acid formation involve the breakdown of hexoses to pyruvate in glycolysis, followed by its decarboxylation to produce acetyl CoA . Very importantly, the CO_2 released during this reaction is not lost, but is recycled by pyruvate carboxylase (produced constitutively in *Aspergillus*) in the anaplerotic formation of oxaloacetate (other anaplerotic routes to oxaloxacetate are also operated). Normally, oxaloacetate would largely supplied through the completion of the TCA cycle, allowing recommencement of the cycle by condensing with acetyl CoA to form citrate, catalysed by citrate synthase.

However, in order to accumulate citrate, its onward metabolism (continuation of the cycle) must be blocked. This is achieved by inhibiting aconitase, the enzyme catalysing the next step in the TCA cycle. Inhibition is accomplished by removal of iron, an activator of aconitase. Consequently, during citrate accumulation, the TCA cycle is largely inoperative beyond citrate formation, hence the importance of the anaplerotic routes of oxaloacetate formation.

Conventional strain improvement methods and the genetic engineering of elements of primary metabolism in *A. niger* are being employed in an effort to improve citric acid production. The aim is to increase metabolic flux leading directly to citric acid formation by decreasing fluxes through branches of this pathway, thus producing fewer byproducts, particularly gluconic acid and oxalic acid. Utilization of mutants lacking glucose oxidase, and consequently unable to produce gluconic acid from glucose, are examples of such an approach. Alternatively, a direct increase of the flux through the main pathway may be achieved by overproduction of the constituent enzymes.

FERMENTATION PROCESSES USED IN CITRIC ACID PRODUCTION

SURFACE AND SOLID-SUBSTRATE FERMENTATIONS

This method use simple technology and have low energy cost, but are more labor intensive. Liquid surface methods involve placing the sterilized medium, usually containing molasses plus various salts, into shallow (5–20 cm deep) aluminium or stainless steel trays stacked in an aseptic room. The medium is formulated with relatively low levels of iron, otherwise the citric acid yield is reduced .The trays are inoculated by spraying with *A. niger* spores, either a spore suspension or dry spores. The fungus then develops on the surface of the medium. Sterile air is blown over these cultures, which is important for maintaining aerobic conditions, temperature control and in lowering the CO_2 level. Medium pH gradually falls to below 2, at which point citric acid production begins. At 30°C, the fermentation takes about 8–12 days to complete and achieves a productivity of around 1.0kg/m_3 per day. Solid-state fermentation process for citric acid production is small-scale operation. Each plant generates only a few hundred tonnes per annum, and uses a solid medium of steam-sterilized wheat bran or sweet potato waste that has 70–80% moisture content. This mash is inoculated with spores of *A. niger* and then spread on trays or a clean floor to a depth of 3–5 cm. Air circulation helps to maintain the temperature at about 28°C. This process runs for 5–8 days, after which the mash is collected and the citrate is extracted using hot water. Various solid food processing residues are being evaluated to determine whether

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they too could serve as low cost substrates for citric acid production. In addition, technological developments are being sought, such as the use of packed bed reactors. Preliminary trials have produced high levels of citric acid with low levels of fungal biomass, as these reactors retard fungal growth and promote greater substrate conversion to citric acid.

SUBMERGED PROCESSES

More than 80% of the worldwide supply of citric acid is produced using submerged batch fermentation in stirred tanks of 40–200m^3 capacity or larger airlift fermenters of 200–900m^3 capacity. The fermenters are corrosion-resistant, made of stainless steel, or steel lined with special glass or plastic.

These fermentations mostly use beet or cane molasses as the carbon source. Unlike surface methods, vegetative inoculum, rather than spores, is normally used. Consequently, the culture organism is taken through several propagation stages in order to generate a sufficient quantity of inoculum. Initially, spores of the production strain of *A. niger* are produced on solid medium and then used to inoculate a small-scale submerged fermentation where fungal pellet formation takes place. Quantities of stable fungal pellets are then developed for the inoculation of the production fermenter. The structure of these pellets has a major influence on productivity. Small pellets of less than 1mm diameter, with fluffy centers and smooth surfaces are preferred.

These structural properties and their physiology are strongly dependent on medium composition and operating conditions. Pellets that produce high levels of citric acid are characterized by short forked, bulbous hyphae. The presence of even low levels of some heavy metals, particularly manganese, can be detrimental to pellet formation, resulting in hyphae that are long and unbranched. Thus, it is necessary to pretreat all raw materials to reduce manganese concentrations to below 0.02 mmol/L. Low manganese levels also limit the operation of the pentose phosphate pathway, which would otherwise divert flux away from glycolysis and reduce citrate production. Alternatively, copper ions may be added to counteract the manganese, by preventing its uptake. Citric acid yields are also improved by formulating the medium with minimum levels of iron. This reduces onward metabolism of citrate because, as mentioned previously, aconitase has a requirement for iron. Also, the addition of copper further diminishes aconitase activity, as it acts as an antagonist to iron, as well as manganese.

In order to maintain good citric acid yields, media sugar concentrations must be at least 140 g/L, which promotes the activity of both glycolytic enzymes and pyruvate carboxylase. It is also important to restrict growth through nitrogen limitation. This is normally accomplished by providing ammonium salts at levels of 0.1–0.4 g/L. The ammonium ions also stimulate citric acid production by counteracting the inhibitory effect of citrate on phosphofructokinase, a key enzyme of glycolysis. These fermentations are highly aerated and maintained at 30°C. For the initial growth phase, the pH starts at 5–7, but must then be kept below 2, otherwise oxalic and gluconic acids accumulate at the expense of citric acid, and low pH inhibits glucose oxidase. Overall yields of 0.7–0.9 g citrate per gram glucose can be attained in these submerged fermentations with productivities of up to 18.0kg/m^3 per day.

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Smaller volumes of citric acid are also produced using yeasts such as *Candida guilliermondii* and *Yarrowia* (formerly *Candida*) *lipolytica*. These yeasts are free from problems with metal ions, and provide shorter and more productive fermentations than those currently available with *A. niger*.

CITRIC ACID RECOVERY

Recovery of citric acid commences with the removal of fungal mycelium from the culture medium. Further polishing filtration may be necessary to remove residual mycelia and precipitated oxalate. The resulting clarified solution is heated and lime (CaO) is added to form a precipitate of calcium citrate. This is separated by filtration and treated with sulphuric acid to generate citric acid and a precipitate of calcium sulphate (gypsum). Following filtration, the dilute citric acid solution is decolorized with activated carbon and evaporated to produce crystals of citric acid. These crystals are recovered by centrifugation, then dried and packaged. Alternate recovery methods being evaluated in order to avoid the use of lime and sulphuric acid include solvent extraction, ion-pair extraction and electrodialysis.

ANTIBIOTICS

Most antibiotics are secondary metabolites produced by filamentous fungi and bacteria, particularly the actinomycetes. Well over 4000 antibiotics have been isolated from various organisms, but only about 50 are used regularly in antimicrobial chemotherapy. The best known and probably the most medically important antibiotics are the β-**lactums**, penicillins and cephalosporins; along with aminoglycosides, such as streptomycin, and the broad-spectrum tetracyclines **.**The remaining fails to fulfill certain important criteria, particularly their lack of selectivity, exhibiting toxicity to humans or animals, or their high production costs. Some antibiotics have applications other than antimicrobial chemotherapy. For example, actinomycin and mitomycin, produced by *Streptomyces peucetius* and *S. caepitosus*, respectively, have roles as anti-tumour agents; and other antibiotics are used for controlling microbial diseases of crop plants, or as tools in biochemistry and molecular biology research. Several antibiotics are also added to animal feed as growth promoters. However, worries about the development of resistance has meant that some antibiotics, used or intended for human use, may be withdrawn from use in animal feed. For example, the EU Commission voted to ban the application of bacitracin, spiromycin, tylomycin and virginiamycin as growth promoters after January 1999.

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B-LACTAMS

Over 100 β-lactams, mostly penicillins and cephalosporins, have been approved for human use, and they account for over half of the antibiotics produced worldwide. This group is especially useful because of their wide margin of safety. They specifically target the synthesis of peptidoglycan, a vital bacterial cell wall component, which is not present in eukaryotic organisms, thus providing a high level of selectivity. They primarily inhibit the cross-linking transpeptidation reaction, resulting in the formation of incomplete peptidoglycan, severely weakening the bacterial cell wall structure.

PENICILLIN

Penicillin was discovered by Fleming in 1928 following his famous observation of an inhibitory zone surrounding a fungal contaminant, *Penicillium notatum*, on a plate of *Staphylococcus aureus*. In the late 1930s Florey, Chain and Heatley characterized the inhibitory compound responsible, penicillin, and developed a protocol that allowed to be produce in a pure form. The discovery of penicillin and its later characterization and purification ultimately led to major advancements in both medicine and fermentation technology. The speed of these developments is greatly influence by the urgent need to supply penicillin during World War II. Penicillin exhibits the properties of a typical secondary metabolite, being formed at or near the end of exponential growth. Its formation depends on medium composition and dramatic overproduction is possible.

However, *P. notatum*, the organism originally found to produce the antibiotic, generated little more than 1 mg/L from the surface cultures initially used for penicillin production. A 20–25 fold increase in yield was achieved when corn steep liquor was incorporated into the fermentation medium. This byproduct of maize processing contains various nitrogen sources, along with growth factors and side-chain precursors, and remains as a major ingredient of most penicillin production media. Even greater penicillin yields were obtained from a closely related species, *Penicillium chrysogenum*, which was originally isolated from a mouldy cantaloup melon. Further increases in yield were achieved when production went over to submerged fermentation. The wartime requirements for penicillin stimulated the rapid development of a large-scale submerged culture system using stirred tank reactors (STRs). Each vessel was continuously stirred via vertical shaft-driven turbine impellers and incorporated air sparging. These technological developments had a major impact on the advancement of the whole field of fermentation technology. Since the 1940s, penicillin yield and fermentation productivity has been vastly improved by extensive mutation and selection of producer strains. The traditional approach to improving penicillin yields involved random mutation and selection of higher producing strains. Resulting mutants were grown in liquid medium and culture filtrates were assayed for penicillin. This was slow and painstaking as large numbers of strains had to be tested. Nevertheless, such methods were the key to the dramatically increased yields achieved since the discovery of penicillin. Penicillin fermentations now produce yields in excess of 50 g/L, a 50 000-fold increase from the levels first produced by Fleming's original isolate.

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The contribution of classical methods of strain improvement has so far outweighed all other approaches, including more recently available genetic manipulation techniques. The latter have contributed more to our understanding of the complex mechanisms of penicillin biosynthesis, particularly the genetic arrangement of improved strains, the identification of bottlenecks in penicillin synthesis and the regulation of secondary metabolism in overproducing strains.

The basic structure of the penicillins is **6-aminopenicillanic acid (6-APA)**, composed of a **thiazolidine ring** fused with a β-**lactam ring** whose 6-amino position carries a variety of **acyl** substituents. This β-lactam– thiazolidine structure, synthesized from **L-aminoadipate**, **L-cystine** and **L-valine**, is common to penicillins, cephalosporins and cephamycins. In the absence of added side-chain precursors to the fermentation medium of *P. notatum* or *P. chrysogenum*, a mixture of natural penicillins is obtained from culture filtrates, notably penicillin G (benzyl penicillin) and the more acid-resistant penicillin V (phenoxymethyl penicillin).

These penicillins are most active against Gram-positive bacteria. However, an expanded role for the penicillins came from the discovery that different biosynthetic penicillins can be formed by the addition of side-chain precursors to the fermentation medium and that natural penicillins can be modified chemically to produce compounds with improved characteristics. Most penicillins are now semi synthetic , produced by the chemical modification of natural penicillins, obtained by fermentation using strains of *P. chrysogenum.* Modification is achieved by removing their natural acyl group, leaving 6-APA, to which other acyl groups can be added to confer new properties. These semisynthetic penicillins, such as methicillin, carbenicillin and ampicillin, exhibit various improvements, including resistance to stomach acids to allow oral administration, a degree of resistance to penicillinase and an extended range of activity against some Gram-negative bacteria.

COMMERCIAL PRODUCTION OF PENICILLIN

Penicillin production is usually via a fed-batch process carried out aseptically in stirred tank fermenters of 40000–200000L capacity, although airlift systems are sometimes used. The fermentation involves an initial vegetative growth phase followed by the antibiotic production phase. Throughout the process, the oxygen level is very important and must be maintained at 25–60mmol/L/h. However, this is not straightforward, because the oxygen transfer rate is affected by the viscosity, which increases as the fermentation progresses. These processes are maintained at 25–27°C and pH 6.5–7.7, the specific conditions depending upon the *P. chrysogenum* strain used.

Various carbon sources have been adopted for penicillin production, including glucose, lactose, sucrose, ethanol and vegetable oils. About 65% of the carbon source is metabolized for cellular maintenance, 25% for growth and 10% for penicillin production. In the past, a mixture of glucose and lactose was used, the former producing good growth, but poor penicillin yields, whereas the latter had the opposite effect. The mode of 'feeding' of a particular carbon source is vitally important, as it can influence the production of this secondary metabolite. Corn steep

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liquor is still being use as a source of nitrogen, additional nutrients and side-chain precursors. Its acidic nature creates a requirement for calcium carbonate (1%, w/v) and a phosphate buffer to neutralize the medium, thereby optimizing its pH for penicillin production. Ammonia, mineral salts and specific side-chain precursors, e.g. phenyl acetic acid or phenoxyacetic acid, may also be added. However, as some precursors are toxic, they must be fed continuously at non-inhibitory concentrations.

Inoculum development is usually initiated by adding lyophilized spores to a small fermenter at a concentration of 5×10^3spores/ml. Fungal mycelium may then be grown up through one or two further stages until there is sufficient to inoculate the production fermenter. Initially, there is a vegetative growth phase devoted to the development of biomass, which doubles every 6 h. This high growth rate is maintained for the first 2 days. To ensure an optimum yield of penicillin in the following production phase, the mycelium must develop as loose pellets, rather than compact forms. During the following production phase, the carbon source is fed at a low rate and penicillin production increases. This continues for a further 6–8 days, provided appropriate substrate feeds are maintained.

Penicillin is excreted into the medium and is recovered at the end of fermentation. Whole broth extraction may be performed, but can lead to downstream processing problems, as additional materials leach from the mycelium. Usually, penicillin recovery follows removal of mycelium using rotary vacuum filters, the efficiency of which may be affected by the culture media composition, particularly its proteinaceous components. Recovered mycelium is then washed to remove residual penicillin, prior to its use as animal feed or fertilizer. Antibiotic recovery is often by solvent extraction of the cell-free medium, which gives yields of up to 90%.

This involves reducing the pH of the filtered medium to 2.0–2.5 by addition of sulphuric or phosphoric acid, followed by a rapid two-stage continuous countercurrent extraction at 0–3°C using amyl acetate, butyl acetate or methyl isobutyl ketene. The low temperature is necessary to reduce damage to penicillin due to the low pH. Alternately, ion-pair extraction may be used at pH 5–7, in which range penicillin is stable. Any pigments and trace impurities are removed by treating with activated charcoal. The penicillin is then retrieved from the solvent by addition of sodium or potassium acetate. This reduces the solubility of the penicillin and it precipitates as sodium or potassium salt. Resultant penicillin crystals are separated by rotary vacuum filtration. Solvent is recovered from the separated liquor and any other materials used, such as the charcoal, which is very important in terms of the overall economy of the process. Penicillin crystals are mixed with a volatile solvent, usually anhydrous ethanol, butanol or isopropanol, to remove further impurities. The crystals are collected by filtration and air dried. At this stage the penicillin is 99.5% pure. This product may be further processed to form a pharmaceutical grade product or is used in the production of semisynthetic penicillins.

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PRODUCTION OF SEMISYNTHETIC PENICILLINS AND CEPHALOSPORINS

As mentioned previously, the objective in semisynthetic penicillin production is to generate compounds with improved properties, e.g. acid stability, resistance to enzymatic degradation, broader spectrum of activity, etc. It involves removal of the side chain of the base penicillin to form 6-APA. This is achieved by passage through a column of immobilized **penicillin acylase**, usually obtained from *Escherichia coli,* at neutral pH. Penicillin G, for example, is converted to 6-APA and phenylacetic acid. The 6-APA is then chemically acylated with an appropriate side chain to produce semisynthetic penicillin.

Yields of cephalosporins from direct fermentations are much lower than those of penicillins. Consequently, as 6-APA can also serve as a precursor of cephalosporins, it is often used as the starting material for their semisynthetic production. A base natural penicillin is converted to 6-APA, as described above, followed by its conversion to the preferred precursor, 7-amino deacetoxycephalosporicacid (7-ADCA), by ring expansion. A suitable side chain can then be readily attached.

EXOPOLY SACCHARIDES

Microbial exopolysaccharides are bio thickeners that can be added to a wide variety of food products, where they serve as viscosifying, stabilizing, emulsifying or gelling agents. Numerous exopolysaccharides with different composition, size and structure are synthesized by lactic acid bacteria. The heteropolysaccharides from both mesophilic and thermophilic lactic acid bacteria have received renewed interest recently. Structural analysis combined with rheological studies revealed that there is considerable variation among the different exopolysaccharides; some of them exhibit remarkable thickening and shear-thinning properties and display high intrinsic viscosities. Hence, several slime-producing lactic acid bacterium strains and their biopolymers have interesting functional and technological properties, which may be exploited towards different products, in particular, natural fermented milks.

EXOPOLYSACCHARIDES FROM FOOD GRADE MICROORGANISMS

Many food grade microorganisms produce EPS, in particular lactic acid bacteria (LAB), propionic bacteria and bifidobacteria. Most of the EPS-producing LAB strains studied in more detail were isolated from dairy products, e.g. Scandinavian ropy fermented milk products, various yogurts and fermented milks and sugary kefir grains. Also cheese and fermented meat and vegetables served as a source of EPS-producing LAB strains.

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SOME OTHER EXOPOLYSACCHARIDE PRODUCERS

Tremella fuciformis, belonging to the order of the Tremellales and the family of the Tremellaceae, has been appreciated as an edible mushroom. It's also been used for medicinal purposes due to its diverse physiological activities such as improving immunodeficiency and preventing senile degradation of micro vessels. All the fractions and sub-fractions of the polysaccharide from fruiting body of *T. fuciformis* displayed the ability to induce human monocytes to produce interleukines (IL-1 and IL-6) and tumor necrosis factor in vitro. Together with this immunomodulatory effect, the acidic heteropolysaccharide lowers the cholesterol level in blood serum.

PIGMENTS

The most commonly used food grade pigments are chemical compounds containing nitrite and nitrate salts. These synthetic compounds have been reported to have carcinogenic and teratogenic effects. This has been one of the major reasons for the increased interest in producing pigments from biological origin like plants and microorganisms .The best known of the microbial colorants are produced by *Monascus* group, specifically *M. ruber, M. anka* and *M. purpureus*, because of their chemical stability. During growth, *Monascus* sp. breaks down starch substrate into several metabolites, of which pigments are produced as secondary metabolites. The structure of pigments depends on type of substrate and other specific factors during culture such as pH, temperature and moisture content.

Monascus fungi produce at least six major related pigments which can be categorized into 3 groups based on color as follows: yellow pigments: monascin (C_{21} H $_{26}O_5$) and ankaflavin (C_{23} H $_{30}O_5$); orange pigments: monascorubrin ($C_{23}H$ $_{26}O_5$) and rubropunctatin ($C_{21}H_{22}O_5$); and red pigments: monascorubramine (C $_{23}H$ $_{27}NO_4$).

In addition, rubropuntamine (C $_{21}H$ $_{23}NO$ $_4$). The color of the pigment is influenced by the culture conditions, pH value and the carbon and nitrogen sources in the substrate *Monascus* pigments can be obtained from both solid state and submerged culture. In Solid state fermentation process, the substrates not only supply the nutrients to the microbial culture growing on it, but also serve as an anchorage for the cells resulting in high pigment productivity. Moreover, it is a low-cost process using agro-industrial residues as substrates. Cheap agricultural products and residues were used as substrates for pigment production such as rice broken, wheat bran, jackfruit seeds, palm kernel cake, cassava starch etc. Since the jackfruit seeds are rich in carbohydrate, protein and trace elements, it can be used as a potential substrate for the production of food grade pigments.

Tips: Most probable questions from this chapter would be on source micro organism used for production of respective products.

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GATE 2011

Q.1 Match the products in Group 1 with their respective organisms in Group 2

Group 1	Group 2
P. Glycerol	1. *Cornyebacterium glutamicum*
Q. Glutamic acid	2. *Alcaligenes faecalis*
R. Curldan	3. *Dunaliella salina*
S. Amphotericin B	4.*Streptomyces nodosus*

(A) P-2, Q-1, R-3, S-4
(B) P-4, Q-2, R-1, S-3
(C) P-3, Q-1, R-2, S-4
(D) P-2, Q-1, R-2, S-4

Answers:

- **Q.1 – C.**

Career Avenues

MICROBIAL PRODUCTION, PURIFICATION AND BIOPROCESS APPLICATION(S) OF INDUSTRIAL ENZYMES

COMMERCIAL MICROBIAL ENZYME PRODUCTION

Microbial enzymes are predominantly produced by submerged fermentations, although some solid-substrate fermentations are used, particularly for the production of extracellular fungal enzymes. In fact, the first commercial microbial enzyme preparation was produced via solid substrate fermentation. This enzyme, '**Takadiastase**', a fungal amylase, was produced by culturing ***Aspergillus oryzae*** on moist rice or wheat bran. The process was initially developed by Dr Jokichi Takamine and patented in the USA in 1884. However, large-scale production of microbial enzymes was not generally feasible until after the middle of the 20th century. It became possible only after the vast improvements to submerged fermentation technology that followed the development of penicillin fermentations in the 1940s. Most industrial enzymes are products of batch processes and few are currently produced via continuous fermentation. The fermenters for bulk enzyme production are up to $100m^3$ capacity, but fine enzymes may be produced on smaller scales of a few hundred litres or less. Most fermenters are stirred tank reactors that are operated under aseptic conditions and use low-cost undefined complex media.

As in the development of any fermentation process, enzyme production processes traditionally begin with the search for a suitable producer organism. Use of GRAS-listed (generally regarded as safe) organisms are an important consideration for enzymes that are to be used in food or medical applications. A programme of microorganism screening and selection is necessary, to determine enzyme properties, such as optimum pH and heat resistance, and examination of the ability to secrete the target enzyme. Enzymes from thermophiles generally provide several advantages. They are thermostable, able to operate at higher temperatures than enzymes of mesophiles, thereby increasing diffusion rates and solubility, and decreasing both viscosity and the risk of microbial contamination. The fermentation system and conditions for maximum production of the enzyme per unit of biomass, using inexpensive carbon and nitrogen feedstocks, must then be determined. Apart from enzyme productivity, a further consideration is enzyme stability, which can influence the timing, and operations used in downstream processing. The level of purification applied varies considerably depending on whether the enzyme is intracellular or extracellular, and on its end use. Downstream processing involves separation, purification, stabilization and preservation Strain improvement may be attempted to further enhance enzyme productivity. In the past, this has often involved cycles of random mutagenesis and screening.

However, other strategies are now available for both organism and protein engineering. Targets for improvement often include increased secretion efficiency for extracellular enzymes and

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overcoming of the organism's own regulatory mechanisms. The latter may involve attempts to relieve catabolite repression, a common regulatory mechanism for many hydrolytic enzymes. Other improvements may be achieved by enhancing mRNA half-life and increasing gene dosage through chromosomal amplification or by plasmid amplification, if the enzyme is plasmid encoded. Targets for enzyme (protein) engineering are enhancement of enzyme activity, improved stability, altered pH optima or temperature tolerance, modified specificity and general improvements to their industrial performance. However, such changes involve the manipulation of specific amino acid constituents and are dependent on prior knowledge of the amino acid sequence of the protein.

Today, if a useful enzyme is identified in a microorganism, plant or animal, which is itself difficult to culture, or if little is known of its physiology and biochemistry, other strategies for commercial production can be adopted. Rather than proceeding with extensive research and development programmes to facilitate the production of the enzyme by the natural producer organism, the structural gene for the enzyme can be transferred into a selected, readily cultivated, 'host' microorganism, along with appropriate mechanisms for control. Genetic engineering now makes possible the expression of the gene coding for almost any enzyme, no matter what its origin. This enables enzyme manufacturers to develop rapidly a process for producing large quantities of the enzyme.

Expression in GRAS-listed organisms provides obvious advantages. Prime candidates for this role are species within three genera of microorganisms, namely *Bacillus*, *Aspergillus* and *Saccharomyces.* Their biology is well understood and they have proved safe to handle, growth is fast and can produce high yields of enzymes, many of which can be excreted into the fermentation medium. A further advantage is that the medium and conditions under which they grow and perform well are already known, thus minimizing further costly experimentation to optimize fermentation conditions.

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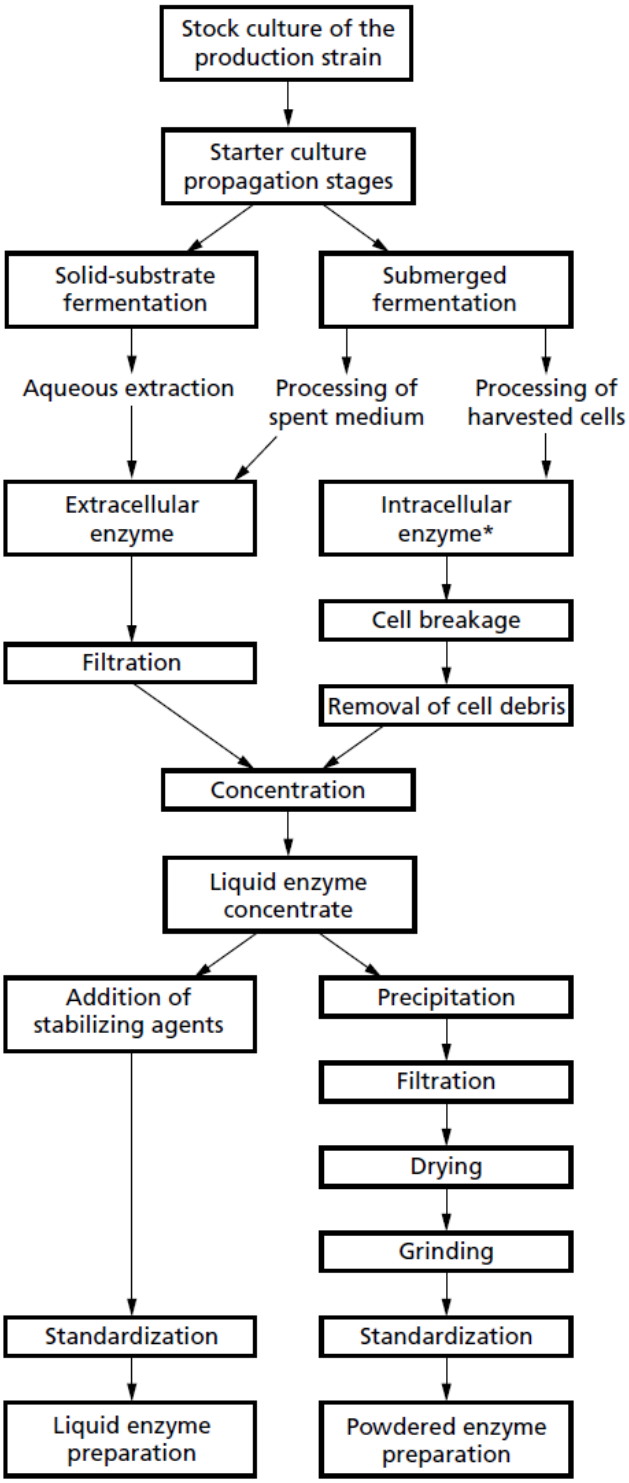


Fig.1. General Process of Industrial enzyme production

DETERGENT ENZYMES

The incorporation of enzymes into detergents provides several benefits. Energy savings are made as a lower wash temperature can be used and levels of less desirable detergent chemicals can be reduced. Unlike other detergent components, enzymes do not have a negative impact on sewage treatment processes. They are totally and rapidly biodegraded to leave no harmful residues. Consequently, they are environmentally safe and are no risk to aquatic life.

The first commercial use of microbial enzymes in detergents was in 1959. A Swiss chemist Jaag, who worked for the detergent company Gebrüder Schnyder, developed a new product containing a bacterial protease, which replaced the animal trypsin that had previously been incorporated into these detergent products. From this point there was an increasing use of microbial enzymes in detergents. In the 1960s, there were further improvements through the introduction of proteases active at alkaline pH. They were relatively unaffected by other components of washing powder and functioned at the desired wash temperatures. A well-known example is subtilisin, a bacterial alkaline serine protease from *Bacillus licheniformis* and *B. subtilis*, which is used extensively in laundry detergents at levels of 0.015– 0.025% (w/w). This enzyme has now been engineered to improve pH and temperature characteristics, and reduce sensitivity to peroxide.

Proteases are not the only enzymes used in detergents, since the late 1980s amylases and lipases have been available for incorporation, e.g. Lipolase from Novo Nordisk. Lipolase was the first detergent enzyme to be produced through recombinant DNA technology. The lipase gene was isolated from a strain of the filamentous fungus *Humicola* and then transferred to *Aspergillus oryzae* which is more readily cultivated in submerged fermentations. Superior fatty acid digesting enzymes have now been discovered, such as a cutinase from *Fusarium solani* which naturally degrades the mixture of fatty acids that form plant cuticles. The gene for this enzyme has been transferred to *Saccharomyces cerevisiae* and commercial production is now being examined.

STARCH PROCESSING ENZYMES AND RELATED CARBOHYDRASES

Microbial enzymes have proved to be of immense value in the processing of starch, a polysaccharide composed of amylose (linear α-1,4-linked glucose units) and amylopectin (branched polymer with both α-1,4 and α-1,6 linkages). α-Amylase (1,4-α-D-glucan glucanohydrolase) is one of the most important of these industrial enzymes. This endo-enzyme acts randomly on α-1,4linkages, and ultimately generates glucose, maltose and maltotriose units. It can also catalyse the cleavage of internal α -1,4 linkages in glycogen and oligosaccharides. Since the 1950s fungal amylases have been used to manufacture sugar syrups containing specific mixtures of sugars that could not be produced by conventional acid hydrolysis of starch. In addition,α-amylases are employed in several industries, particularly brewing and baking, where starch liquefaction and dextrin hydrolysis are required. Many other microorganisms, including bacteria and some yeast, also secrete them. The thermostable **α -amylases** from *Bacillus* species, e.g. *B. licheniformis*, have proved to be especially useful.

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A further success has been **amyloglucosidase (glucoamylase)** from ***Aspergillus niger*** and ***Rhizopus*** species, which first became available in the early 1960s. This enzyme can completely break down starch and dextrins into glucose. Within a few years of its introduction, almost all conventional glucose production via acid hydrolysis was replaced by enzyme-based processes, because of the greater yield, higher degree of purity and easier product crystallization. The process was further improved by using highly thermostable bacterial amylase in starch pretreatment (liquefaction).

Glucose isomerase has also been a notable success in the starch processing industry. This enzyme can be obtain from many bacteria, including species of ***Bacillus*** and ***Streptomyces***, and is usually immobilize for use in the conversion of glucose to fructose. The product where conversion to fructose is completed has the same calorific value as glucose but its sweetening effect is approximately high as twice. Substrate glucose is normally derived from hydrolysed corn starch and is then converted to a mixture of glucose and fructose by glucose isomerase. Practical conversion yields are in the range of 40–50% and the products are referred to as 'high fructose corn syrup' (HFCS) or 'isosyrup'. Fructose concentration can be increased to 55% by chromatographic enrichment. Alternatively, the fructose can now be separated from the glucose–fructose mixture and the remaining glucose fraction is isomerized to produce a final combined product containing up to 80% fructose. These syrups derived from starch are somewhat cheaper than sucrose from cane or beet source but have the same sweetening power. Consequently, they have become major food ingredients particularly in North America and annual production of HFCSs is now excess of 8×10^9 kg.

Invertase (β-fructofuranosidase) was the first enzyme to be immobilized for use on an industrial scale. It was developed in the UK by Tate and Lyle during the early 1940s for syrup production. The conversion of sucrose to glucose and fructose using this immobilized invertase replaced conventional acid hydrolysis method. This enzyme was originally prepared from autolysed *S. cerevisiae* preparations, which were adjusted to pH 4.7, clarified by filtration through calcium sulphate and then adsorbed onto bone char. Invertase is obtained from other yeasts and filamentous fungi e.g. *A. niger* and *A. oryzae*. Apart from syrup production it is employed in confectionery manufacture. For example soft centred chocolates are prepared by taking a solid sucrose-based filling containing some invertase and coating with chocolate. Within 2 weeks the centre becomes converted into fructose/glucose syrup.

Lactase (β-galactosidase) is employed in several industrial processes that require the hydrolysis of lactose from milk where the disaccharide is present at a concentration of 4.7% (w/v). Hydrolysis of lactose to glucose and galactose is performed on milk and milk products for infants who are lactose-intolerant and in regions of the world where a large proportion of the adult population exhibit lactose-intolerance e.g. regions of Africa and South-East Asia. Lactose hydrolysis is also useful in the manufacture of products such as ice cream as the low solubility of this disaccharide leads to crystal formation that may give an unpleasant sandy texture. In addition its conversion to glucose and galactose increases the relative sweetness by about four-fold.

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Commercial lactase is obtained from ***Kluyveromyces marxianus*** (formerly *K. fragilis*), ***A. niger*** or ***A. oryzae***. The yeast enzymes have optimum pH around 6-7 whereas the *Aspergillus* enzymes have low pH around 3-4. These enzymes may be used free or immobilized the latter being particularly useful for whey treatment. Yeast enzymes have been immobilized by incorporation into cellulose acetate fibers and the *Aspergillus* enzymes have been immobilized on 0.5mm diameter porous silica for use in packed bed reactors.

ENZYMES IN CHEESE PRODUCTION

Rennet preparations from the stomachs of calves, lambs and kids have been used in cheese production for thousands of years. These rennets contain the enzyme rennin (chymosin an aspartic protease) which performs limited proteolysis of milk protein (casein) to form curds at the start of cheese making. Fig sap containing the proteolytic enzyme ficin has sometimes been used for the same purpose but many other proteases carry out too much proteolysis.

Animal rennets particularly from calves predominated in cheese manufacture until 1960s when increase in cheese production coincided with a shortage of available animal rennet. Specific fungal proteases which have very similar properties to calf chymosin were then developed as microbial rennets such as proteases from *Rhizomucor miehei* and *R. pusillus*. This overcame the shortfall and facilitated the production of so-called 'vegetarian' cheese. Some microbial enzyme preparations are also more temperature-sensitive (thermolabile) than calf rennet which is useful in certain cheese-making processes. More recently calf chymosin gene has been introduced into several microorganisms, including *E. coli*, *K. lactis*, *Aspergillus nidulans* and *A. niger*var.

These genetically engineered microorganisms are capable of expressing and secreting the enzyme. Recombinant chymosin preparations now have approval for use as food additives in over 20 countries. Animal and microbial chymosins (constituting about one-third of the market) have combined world sales worth in excess of $75 million. Microbial lipases are also used in dairy products especially cheese for the hydrolysis of fatty acid esters to accelerate flavour development.

ENZYMES IN PLANT JUICE PRODUCTION

Several microbial enzymes are employed in fruit juice processing the most important are pectinases. These are often produced by solid-substrate fermentation of *A. niger* or *Penicillium* species. Fruits and berries contain varying amounts of pectin which acts as a binding layer between plant cells to hold adjacent cell walls together. Pectin is a heteropolymer of galacturonic acid, methyl esters of galacturonic acid and other sugar residues. In plant juice production some of this pectin is extracted during pressing. It causes an increase in juice viscosity leading to difficulties in obtaining optimal juice yields, juice clarification and filtration. These problems can be overcome by adding pectinase preparations to the fruit pulp before pressing. Similar enzyme treatment is used to increase the yields of oils from olive pulp, palm fruit and coconut flesh.

Career Avenues

The commercial pectinase is not one enzyme but is usually a complex cocktail of enzymes (pectin methyl esterases, polygalacturonases and pectin lyases) capable of attacking a variety of bonds in correspondingly diverse pectin molecules. Compositions of commercial pectinase preparations vary considerably in the proportions of these different enzymes. Fungal pectinases are also available that remain active in acidic juices from citrus fruits where the pH can be as low as 2. They may be used in the enzymatic peeling of citrus fruit for canning.

The polysaccharide araban a polymer of the pentose arabinose is also an important component of fruit cell walls. Like pectin it is often extracted during pressing of some fruits especially pears. This may lead to haze formation but it can now be eliminated by using commercial arabanases. Also some extracts such as apple juice contain starch which must be degraded to produce clear juices and concentrates. This is achieved by adding amylases along with the pectinase during depectinization of the juice.

In wine making commercial enzymes are used for several purposes and as well for juice extraction. For red wines colour extraction from the grape skins during pressing can be promoted by the addition of commercial cellulases from the fungus *Trichoderma reesei*. In addition glycosidases can be employed to hydrolyse terpenyl glycosides releasing the terpenes that are important constituents of the wine bouquet. Wines prepared from grapes allowed to undergo attack by the fungus *Botrytis cinerea* before harvesting are often difficult to clarify and filter due to the presence of fungal β-glucans. These polymers can be degraded by adding a specific microbial β-glucanase to the wine. Citrus fruits such as grapefruit and bitter oranges contain the bitter-tasting fruit flavonoid called naringin. The bitterness of products derived from these fruits can be adjusted using naringinase (α-rhamnosidase + β-glucosidase) from *A. niger*. It first converts naringin to the less bitter compound prunin then to non-bitter naringenin. The level of naringin hydrolysis may be controlled by regulating the flow rate of fruit juice through a column of immobilized naringinase. Glucose oxidase from *A. niger* or *Penicillium* species are often coupled with catalase which can be employed to remove molecular oxygen from wine, beer, fruit juices and soft drinks. This prevents potentially damaging oxidation which otherwise affects product quality.

$$2\,\text{glucose} + O_2 \xrightarrow{\textit{glucose oxidase + catalase}} 2\,\text{gluconic acid}$$

ENZYMES IN TEXTILE MANUFACTURE

Enzymes are being increasingly used in textile processing for the finishing of fabrics and garments especially in desizing, biopolishing and denim washing. Prior to weaving fabrics from cotton/ blends of cotton and synthetic fibres the threads are reinforced with an adhesive (termed 'size') to prevent breakage of warp threads. The size may be composed of starch, starch derivatives, vegetable gum and water-soluble cellulose derivatives such as methyl and carboxymethylcellulose. Before dyeing, bleaching and printing the size must be removed from the cloth. This has previously been accomplished by treatment with acids, alkalis and oxidizing agents that may damage the fibers. Consequently enzymatic desizing with amylases or cellulases

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is now widely used as they are non-corrosive and produce no harmful effluent wastes. Cellulases are also employed in bio-polishing cotton and other cellulose fibers to produce fabrics with a smoother and glossier appearance. Similar processes using microbial proteases have been developed for treating wool fibers which are composed of keratin. Treatment of denim garments to give a worn look before sale is called 'stonewashing'. Traditionally this has been accomplished by washing the garments in a tumbling washing machine with pumice stones. Cellulases are now used to accelerate abrasion and aid the loosening of the indigo dye. This 'bio-stonewashing' is less damaging to the garments, reduces wear on machines and generates less pumice dust.

ENZYMES IN LEATHER MANUFACTURE

Proteases and lipases are now extensively used in the processing of hides and skins. These enzymes are easier to use, more pleasant to handle and safer than the harsh chemicals that were previously employed. Their most important applications are in soaking, dehairing, degreasing and baiting. Soaking is the first important operation of leather processing. Apart from cleaning hides and skins by removing debris derived from blood, flesh, grease and dung, it rehydrates them. Proteases enhance water uptake by dissolving intra fibrillary proteins that cement the fibers together and prevent water penetration. Lipases are also used to disperse fats as an alternative to degreasing with solvents. These enzymes hydrolyse surface fats and those within the structure of the hides and skin without causing any physical damage.

Dehairing of hides and skins traditionally employed chemicals such as slaked lime and sodium sulphide which in past made tannery effluent a severe pollution problem. Enzyme-assisted dehairing involves proteases usually alkaline protease from *Aspergillus flavus*. Their use reduces or totally replaces the requirements for chemical processing and provides a major cost reduction in effluent treatment. Leather baiting prepares the leather for tanning. It involves removal of any remaining unwanted protein to make the grain surface of the finished leather clean, smooth and fine. Traditional methods employed dog, pigeon or chicken manures. These were very unpleasant to use, unreliable and slow which has been replaced by microbial proteases.

ENZYMES USED IN THE TREATMENT OF WOOD PULPS

Paper manufacture is a major industry. In the USA alone over 70 million tonnes of paper and paperboard are manufactured each year which have a value in excess of US$50 billion. Paper manufacture involves the processing of mostly virgin fibers but there is increasing utilization of some recycled materials or secondary fibers. Recycling saves both trees and energy, decreases waste effluents and reduces landfill loads. Traditionally wood pulp processing involved the extensive use of chemicals which can lead to problems with effluent treatment and environmental pollution. Thus, the development of enzyme-based technology for pulp processing and paper manufacture has major advantages.

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Microbial enzymes can be used in several stages of pulp and paper processing to:

- enhance pulp digestion;
- improve drainage rate removal in water during paper formation;
- increase fiber flexibility;
- selectively remove xylan without affecting other components;
- remove resins;
- enhance bleaching;
- remove contaminants, such as de-inking of high-quality waste paper; and
- Fibrillate or increase interfiber bonding in chemical pulps and herbaceous fibers.

Those microbial enzymes used in these processes include a wide range of cellulases, hemicellulases, pectinases and lipases. However, they cannot yet replace all mechanical and chemical treatments.

ENZYMES AS CATALYSTS IN ORGANIC SYNTHESIS

There is a rapidly growing market for microbial enzymes used in the synthesis of high-value organic compounds for the chemical, food and pharmaceutical industries. One commercially valuable area involves the application of cyclodextrin glycosyltransferases, usually obtained from extreme thermophiles to create cyclodextrins from simple starch molecules. These cyclodextrins are useful carriers of fragile compounds such as vitamins and flavours. No chemical means exists for creating this class of substances so the enzymatic route is unique.

An increasingly important field is the synthesis of chiral compounds. Chiral compounds exist in two forms called enantiomers which are non-superimposable mirror images of each other and are therefore asymmetrical; virtually in all respects the two enantiomers are physically and chemically identical. However, solutions of one enantiomer rotate the plane of polarized light in a clockwise direction (+), whereas solutions of the other enantiomer rotate it anticlockwise (-) by exactly the same amount. This phenomenon is known as optical activity and enantiomers are referred to as optical isomers.

Chirality is vitally important in biology as most natural organic compounds are chiral. For example most amino acids are enantiomeric with only one enantiomer usually being found in nature. However, when the compounds are chemically synthesized an almost 50: 50 mixture of enantiomers are made which is called a racemic mixture. Usually, for most biological molecules, only one of its variants or enantiomers is biologically effective as an enzyme substrate, pharmaceutical drug, nutrient, etc. Its other enantiomer may be useless/futile or even damaging to health. The reason that identical molecules of opposite chirality can have such different biological behaviour is that proteins, especially enzymes and nucleic acids are themselves chiral

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molecules. Consequently, any drug or nutrient must have the proper chirality if it is to interact with a protein of specific chirality. In the industrial synthesis of such compounds, there should be controls to guarantee the production of only the desired enantiomer often at purity higher than 99%. Not only is it economically disadvantageous to make a large quantity of chemicals when only half (one enantiomer) is biologically functional, but for some drugs the other enantiomer may be toxic. Certain chiral syntheses can be performed chemically but enzymatic routes are usually preferred.

Enzyme-based processes can be used to prepare specific enantiomers and resolve enantiomers from racemic mixtures. These processes utilize racemases, which are a family of enzymes that accept either enantiomer, although they are themselves chiral like all other proteins. Their biological function is to turn a substrate of one chirality into its opposite form. They can be used to synthesize chiral amines, alcohols and many other compounds.

GATE 2010

Q.2 Identify the enzyme that catalyzes the following reaction ?



(A) Glutamate synthase (B) Glutamate oxoglutarate aminotransferase
(C) Glutamate dehydrogenase (D) α- ketoglutarate deaminase

Q.3 Match the products in Group I with the microbial cultures in Group II used for their industrial production .

Group I	Group II
P. Gluconic acid	1. *Leuconostoc mesenteroids*
Q. L- Lysine	2. *Aspergillus niger*
R. Dextran	3. *Brevibacterium flavum*
S.Cellulase	4. *Trichoderma reesei*

(A) P-2, Q-1, R-3, S-4 (B) P-1, Q-3, R-4, S-2
(C) P-2, Q-3, R-1, S-4 (D) P-3, Q-2, R-4, S-1

GATE 2012

Q.4 Determine the correctness or otherwise of the following **Assertion** (a) and **Reason** (r).

Assertion: Cell mass yield of a methylotrophic yeast is more on methanol compared to glucose.
Reason: Methanol has a greater degree of reductance compared to glucose.

(A) Both (a) and (r) are correct and (r) is the correct reason for (a)
(B) (a) is correct, (r) is false
(C) (a) is false, (r) is correct
(D) Both (a) and (r) are correct but (r) is not the correct reason for (a)

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Answers:

- **Q.2 – C.**
- **Q.3 – D.**
- **Q.4 – A.**

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PRODUCTION AND PURIFICATION OF RECOMBINANT PROTEINS ON A LARGE SCALE

PROCESSES OF RECOMBINANT DNA TECHNOLOGY

Recombinant DNA technology involves several steps in specific sequence such as isolation of DNA, fragmentation of DNA by restriction endonucleases, isolation of a desired DNA fragment, ligation of the DNA fragment into a vector, transferring the recombinant DNA into the host, culturing the host cells in a medium at large scale and extraction of the desired product. Let us examine each of these steps in brief.

ISOLATION OF THE GENETIC MATERIAL (DNA)

Recall that nucleic acid is the genetic material of all organisms without exception. In majority of organisms this is deoxyribonucleic acid or DNA. In order to cut the DNA with restriction enzymes it needs to be in pure form, free from other macro-molecules. Since the DNA is enclosed within the membranes, we have to break open the cell to release DNA along with other macromolecules such as RNA, proteins, polysaccharides and also lipids. This can be achieved by treating the bacterial cells/plant or animal tissue with enzymes such as **lysozyme** (bacteria), **cellulase** (plant cells), **chitinase** (fungus). You know that genes are located on long molecules of DNA interwined with proteins such as histones. The RNA can be removed by treatment with ribonuclease whereas proteins can be removed by treatment with protease. Other molecules can be remove by appropriate treatments and purified DNA ultimately precipitates out after the addition of chilled ethanol. This can be seen as collection of fine threads in the suspension.

CUTTING OF DNA AT SPECIFIC LOCATIONS

Restriction enzyme digestions are performed by incubating purified DNA molecules with the restriction enzyme at optimum conditions for that specific enzyme. Agarose gel electrophoresis is employed to check the progression of a restriction enzyme digestion. DNA is a negatively charged molecule hence it moves towards the positive electrode (anode). The process is repeated with the vector DNA also. The joining of DNA involves several processes. After having cut the source DNA as well as the vector DNA with a specific restriction enzyme, the cut out 'gene of interest' from the source DNA and the cut vector with space are mixed and ligase is added. This results in the preparation of recombinant DNA.

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AMPLIFICATION OF GENE OF INTEREST USING PCR

PCR stands for **Polymerase Chain Reaction**. In this reaction, multiple copies of the gene (or DNA) of interest is synthesised *in vitro* using two sets of primers (small chemically synthesised oligonucleotides that are complementary to the regions of DNA) and the enzyme DNA polymerase. The enzyme extends the primers using the nucleotides provided in the reaction and the genomic DNA as template. If the process of replication of DNA is repeated many times, the segment of DNA can be amplified to approximately billion times. Such repeated amplification is achieved by the use of a thermostable DNA polymerase (isolated from a bacterium, *Thermus aquaticus*), which remain active during the high temperature. The amplified fragment if desired can now be used to ligate with a vector for further cloning.

INSERTION OF RECOMBINANT DNA INTO THE HOST CELL/ORGANISM

There are several methods of introducing the ligated DNA into recipient cells. Recipient cells after making them 'competent' to receive, take up DNA present in its surrounding. So, if a recombinant DNA bearing gene for resistance to an antibiotic (e.g., ampicillin) is transferred into *E. coli* cells, the host cells become transformed into ampicillin-resistant cells. If we spread the transformed cells on agar plates containing ampicillin only transformants will grow untransformed recipient cells will die. Since, due to ampicillin resistance gene, one is able to select a transformed cell in the presence of ampicillin. The ampicillin resistance gene in this case is called a **selectable marker.**

OBTAINING THE FOREIGN GENE PRODUCT

When you insert a piece of alien DNA into a cloning vector and transfer it into a bacterial, plant or animal cell, the alien DNA gets multiplied. In almost all recombinant technologies, the ultimate aim is to produce a desirable protein. Hence, there is a need for the recombinant DNA to be expressed. The foreign gene gets expressed under appropriate conditions. The expression of foreign genes in host cells involves understanding many technical details.

After having cloned the gene of interest and having optimised the conditions to induce the expression of the target protein one has to consider producing it on a large scale. If any protein encoding gene is expressed in a heterologous host, is called a **recombinant protein**. The cells harbouring cloned genes of interest may be grown on a small scale in the laboratory. The cultures may be used for extracting the desired protein and then purifying it by using different separation techniques.

The cells can also be multiplied in a continuous culture system wherein the used medium is drained out from one side while fresh medium is added from the other to maintain the cells in their physiologically most active log/exponential phase. This type of culturing method produces a larger biomass leading to higher yields of desired protein. Small volume cultures cannot yield appreciable quantities of products. To produce in large quantities the development of **bioreactors** where large volumes (100-1000 litres) of culture can be processed was required. Thus, bioreactors can be vessels in which raw materials are biologically converted into specific products, individual enzymes, etc. using microbial, plant, animal or human cells. A bioreactor

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provides the optimal conditions for achieving the desired product by providing optimum growth conditions (temperature, pH, substrate, salts, vitamins, oxygen).

CHARACTERIZATION OF PURIFIED RECOMBINANT PROTEINS

The absolute purity of a biological substance is hard if at all possible to determine regular and sometimes only subtle protein modifications such as glycosylation, alternative disulphide bond formation, deamidation, oxidation, phosphorylation, acetylation, sulfation, sulfoxidation, γ -carboxylation, and pyroglutamate formation lead to protein variants that may have more or less different characteristics. Also, truncated protein variants might be generated by the presence of cryptic or alternative start sites of transcription by premature stop of the peptide chain elongation process or by the action of host cell peptidases. Peptide mapping and mass spectrometry (MS) usually achieve detection of most of such protein variants. Aggregation is another modification of a protein which can be the result of glycosylation, oxidation, and/or deamidation and can be detected by size-exclusion chromatography. The amount of aggregated protein usually should stay below 5%. It is highly recommended to investigate the nature and potential toxicity of such alterations.

To analyze these variants is an essential yet challenging task as their physicochemical features might not be very different from each other. Owing to the possible presence of highly related protein variants in the preparation it is recommended to determine the purity of a protein drug by at least two independent methods that is methods that use different physicochemical principles such as SDS-polyacrylamide gel electrophoresis and reverse-phase high pressure liquid chromatography (HPLC). Besides these protein variants, so called process related impurities have to be considered.

Of major concern are residual antibiotics from fermentation, enzymes and antibodies from chromatography columns and other column leachates, endotoxin from bacterial hosts, (retro-) viruses, bacteria, fungi, mycoplasma, prions, and various other media components such as solvents, antifoam agents, heavy metal ions, as well as preservatives and expression host components. For DNA contaminations, less than 10 to 100 pg per dose are allowed in the final drug product. Moreover specific impurity standards should be used for impurity quantification and the limit of detection/quantification (LOD/Q) for impurity assays should be indicated. The acceptance limits should not be set higher than safety data and it should not be lower than what is historically achievable by the manufacturing process and by reasonable analytical efforts. In some instances the protein drug is conjugated to effector functions such as radioisotopes, toxins, or other proteins such as cytokines that mediate a biological effect. Besides considering all aspects mentioned above and below for the individual components of the conjugate, special care has to be taken to determine the average coupling ratio as well as the amount of free components, if any in the preparations.

IDENTITY

Identity assays aim at confirming the molecular composition and, if technically possible and commercially reasonable, structure of the drug substance, and thus should be suited to allow the detection of even minor alterations in the molecular composition of the drug. Amino acid

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analysis and peptide mapping are standard methods in the course of protein identification processes.

Molecular mass determination of whole molecules as well as peptide fragments with accuracies of about 0.01% by either matrix-assisted laser desorption/ ionization (MALDI)-MS for surface immobilized samples, or electrospray ionization (ESI)-MS for liquid samples, is another highly efficient protein identification method. These methods additionally support the identification of posttranslational modifications such as glycosylation, glycation, phosphorylation, sulfation, etc. Besides amino acid analysis and elaborated mass spectroscopy techniques, many more analytical methods are applied to support the identity examinations of the protein drug; such as determination of the extinction coefficient, isoelectric point, and crystal structure, as well as recording the nuclear magnetic resonance (NMR) and circular dichroism (CD) spectra and determining the chromatographic profiles from HPLC-runs as well as from capillary and polyacrylamide gel electrophoresis (CE and PAGE, respectively). Automated systems as well as microfluidic devices (‘‘lab-on-the-chip’’) for chromatographic separation of proteins and their subsequent analysis have a huge potential to dominate the analytics field in the future.

QUANTIFICATION OF THE PROTEIN

Many physicochemical assays are established to quantify the protein mass. It is determined by exploiting the extinction coefficient in optical density measurements or by colorimetric assays such as the Bradford, Lowry, bicinchoninic (BCA) and biuret assay. Albeit easy to perform, these colorimetric assays suffer from inaccuracies that are due to the use of inappropriate standards like bovine serum albumin. If relevant standards are not available, quantitative amino acid analysis, the (micro-) Kjeldahl nitrogen method or gravimetry are very accurate methods but time consuming alternatives can be applied. Bioassays and immunoassays are also exploited to quantify the protein amount, which have to be validated for accurate measurements and definitively require a reference standard.

STABILITY, STORAGE, AND STERILITY

Stability studies include the evaluation of those protein features that are susceptible to change during storage and might influence the quality, safety, and efficacy. The testing should cover physicochemical, biological, and microbiological aspects, as well as the preservative content such as antioxidants or antimicrobials. Stability-indicating assays are validated quantitative analytical procedures that can detect drug alterations over time. They should include tests for integrity of the drug, potency, sterility, and if applicable moisture, pH, and preservative stability measured at regular intervals throughout the dating period. One key parameter for stability testing is temperature. Real-time stability studies may be confined to the proposed storage temperature. Accelerated stability tests can be conducted at elevated temperatures exceeding standard storage temperatures. Data from accelerated stability studies are supportive but do not substitute for real-time data. They may help to validate the respective analytical assays to elucidate the degradation profile of the drug and to assess the drug stability under storage conditions other than those proposed. Evaluation of the effect of humidity on drug stability may be omitted if the drug container gives appropriate protection against variations in humidity.

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Other parameters if indicated have to be considered for stability testing are the effect of light exposure, container/closure drug interactions, and stability after reconstitution of a freeze-dried product. Sterility testing should be performed initially and at the end of the proposed shelf life.

Besides stability and sterility, other characteristics of the sample such as visual appearance (color, opacity and particulates), dissolution time, and osmolality also have to be described for the drug product in its final container. Although the biological drug may be subject to substantial losses of activity over storage time, the regulatory authorities have provided little guidance concerning release and end of shelf-life specifications. It thus remains a case-by-case decision whether the loss of activity is considered acceptable.

SPECIFICITY AND CROSS-REACTIVITY

Assays have to be designed for the evaluation of specificity of the given drug-target interaction especially when considering antibodies as a very important class of protein drugs. One may distinguish three types of assays for specificity assessment:

- Binding assays, which include appropriate positive and negative controls as well as target molecule control that ideally consists of the closest target-related variant(s). Preferably quantitative inhibition assays are performed with soluble target preparations which distinctly enhance the confidence in the specificity of the drug-target interaction.
- Determination of the molecular nature of the drug binding site by epitope mapping or by measuring the impact of carbohydrates on the binding site and of other modifications that might modulate the binding event.
- Immunohistochemistry (IHC) to scan for cross-reactivity with human (and animal) tissues.Quick-frozen surgical samples is preferred over postmortem tissue samples. Tissues from at least three unrelated human donors should be evaluated. If with these assays cross-reactivity in vitro to non-target molecules or tissues is detected, then testing in vivo for crossreactivity in animal models if available is indicated.

POTENCY DETERMINATION GENERAL ASPECTS AND ASSAYS

Protein therapeutics often exerts an exquisite biological function at very low concentrations in the pico- to nanomolar range. Sensitive methods for the determination of the biological activity have to be established that reliably allow the quantification of this prominent feature of protein drugs. It is of atmost important to define a reference standard in quantitative terms which is difficult undertaking for protein drugs. Owing to microheterogeneities denatured inactive material, various a glycosylation pattern, and quite different set-ups of potency assays, remains an elusive challenge to define appropriate standards. Applying the mildest purification schemes starting from the natural protein source (and not from recombinant expression systems), using appropriate storage conditions to keep the protein in its active conformation, as well as robust and sensitive potency assay conditions are the essentials for preparation of a reference standard. With such reference in hand the potency is routinely expressed in activity units per milligram of pure protein. The WHO is providing calibrated potency standards that are available for

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commercial and academic organizations. If there is no official source a reference standard should be of the highest possible purity that can be obtained with reasonable efforts and should be fully characterized as described above. For antibodies the prominent biological function is binding to the antigen. Besides the manifold antibody binding assays isothermal titration calorimetry (ITC) could become an elegant means for reference standard evaluation. The fraction of binding antibody in the sample is determined by measuring the stoichiometry of the antigen–antibody interaction. For example with a monovalent Fab antibody fragment a 1 : 1 stoichiometry for binding to a monomeric antigen is expected, if all Fab molecules are in their active binding conformation. For correct interpretation of data, antibody-ITC relies on the presence of homogenous epitope presenting antigen preparations and on not too high affinities of the interactions. In the case that a protein exerts more than one biological activity the assay that most closely reflects the clinical situation should be chosen. Assays for measuring further biological function of the drug should also be established to assess possible side effects in vivo. However, two protein preparations with identical specific activity units or functionality in vitro may behave very differently in vivo. For example, different glycosylation patterns may lead to very similar or even identical molecular masses and potency in vitro but the serum half-life and immunogenicity of both variants may significantly vary from each other and thus may lead to distinctly different effects in vivo.

In general, potency determination methods can be grouped into (1) biochemical assays that use defined reagents in vitro, (2) and into cell-based (3) as well as animal based assays that rely on living systems.

BIOCHEMICAL ASSAYS

In most biochemical assays the specific binding of the protein drug to its target molecule is determined. The target can be either the native or recombinant protein or cells expressing the target molecules. Typically, the law of mass action governs these assays. Today's most often exploited binding assay is the immunoassay that relies on antibodies as detection reagent. Historically, the radioimmunoassay (RIA), which uses radioisotope labeling to track the binding event paved the way to most of the nonradioactive immunoassays such as the outstanding enzyme-linked immunosorbent assay (ELISA) and its many variations for example, sandwich ELISA, capture ELISA, inverted ELISA, and competitive ELISA. Besides immunoassays receptor-binding assays and substrate-binding assays are also commonly applied.

The quality of any immunoassay mainly depends on the specificity, affinity, and stability of the used antibody reagents. Binding of the antibody reagent to the protein drug yields a signal via some enzyme, fluorescence, luminescence or radioisotope label on the detector antibody.

This signal is then transformed into binding constants (e.g. IC_{50} values) and/or concentrations. Another very elegant binding assay is based on surface plasmon resonance (SPR) measurements. It is a label-free method in which one reaction partner is immobilized on a chip, while the other reactant in solution is flowing over the chip. Upon binding, an optical signal is generated and

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recorded as a function of time. With this fast method high and low affinity interactions are detected with only minute amounts of sample. Hence, surface plasmon resonance determination has become a very attractive technique not only for comprehensive evaluation of the binding event but also for the detection of anti-drug antibodies in the sera of patients. In general, biochemical assays are cheap and easy to perform and allow high throughput binding measurements at high accuracy and precision. On the other hand, the biological activity of the protein drug cannot be appropriately assessed by such assays since; the biologically inactive fraction of the drug is often detected as well. For antibodies, biochemical assays are key method for evaluation of their binding characteristics but antibody potency assays should also consider the anticipated mode of action of the therapeutic antibody, beyond the mere antigen-binding event, for example, induction of signal cascades that trigger cell proliferation or cell death.

CELL-BASED ASSAYS

The basis for setting up cell-based assays for potency determination is the availability of responsive cells that are either immortalized, freshly isolated from tissues, or generated by engineering to obtain heterologous expression of the target molecule of choice.

The typical read outs are cell proliferation, differentiation, apoptosis, cytotoxicity (e.g. for antibodies: ADCC - antibody dependent cell-mediated cytotoxicity) and CDC (complement-dependent cytotoxicity), chemotaxis, signal transduction, secretion of biologically active substances, or reporter gene approaches relying on green fluorescent protein or luciferase constructs. Owing to the complex reactive responses of cells, the thorough validation of cell-based assays is highly recommended for a meaningful interpretation of data. For example, the addition of drug-neutralizing compounds to the assay is recommended as a check for specificity. Critical aspects of cell-based assays include the cell immortalization method, cell culture history, passage number, stability of cell line, mycoplasma infection, surface marker pattern, and the effect of fetal calf serum in the medium.

ANIMAL-BASED ASSAYS

Animal-based assays are briefly described here as potency assays, and not as in vivo proof of principle studies in preclinical development. Animals, in particular non-human primates, as well as tissues and organs from animals provide a metabolizing environment and thus offer the advantage of being closer to the ‘‘real-life’’ situation in patients. Especially, bioavailability and toxicity aspects are thus integrated in animal-based assays. However, the exploitation of animal-based potency studies are significantly hampered by the fact that they only provide a low throughput at high costs, take long time while yielding high variability in the results and finally raise serious ethical questions. Hence, animal-based assays are rarely used for potency.

POTENCY DETERMINATION IN BIOLOGICAL MATRICES

Besides potency assessment of the purified drug, potency assays are also applied for quantification of the protein activity in biological matrices from animals/humans. Since the

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concentration of the therapeutic protein in the biological sample is generally low (pico- to nanomolar), standard chromatographic methods to enrich the protein prior to analysis are often not applicable. Hence, the functionality of the protein has to be determined in a complex matrix of other accompanying unrelated (or sometimes even related) proteins, non-protein macromolecules, and low molecular weight compounds such as salts and colored ingredients. During the past decade, many assays have been devised for measuring the potency of proteins under these conditions. By spiking in the purified drug as well as the reference standard into the matrix and by testing series of dilutions with the matrix as diluent, the matrix effect can be assessed and adequately considered in the calculations.

Special focus has to be directed towards the interference with endogenous native, drug-related protein(s) in the matrix that have to be subtracted from the signal. In general, the storage of biological samples at -80^0C in aliquots is indicated in order to avoid degradation events. Consequently, the thorough validation of potency assays, as well as their refinement from the very beginning of drug development, is of key importance.

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CHROMATOGRAPHIC AND MEMBRANE-BASED BIOSEPARATION METHODS

CHROMATOGRAPHY

Chromatographic techniques are usually employed for higher-value products. These methods, normally involving columns of chromatographic media (stationary phase) are used for desalting, concentration and purification of protein preparations. In choosing a chromatographic technique a number of considerations must to be taken into account. For protein products these factors include molecular weight, isoelectric point, hydrophobicity and biological affinity. Each of these properties can be exploited by specific chromatographic methods that may be scaled up to form an industrial unit process.

The order and choice of technique will depend on the particular product but the following chromatographic parameters should be considered: **capacity**, **recovery** and **resolving power (selectivity)**. The capacity refers to the sample size that can be applied to the system in terms of protein concentration and volume and the recovery is the yield of product at each stage. Yield values should be as high as possible otherwise the overall process will be uneconomic. Resolving power and selectivity relate to the ability to separate two components, one being the product and the other the impurity. This is particularly important at the final purification stage. Care must be taken not to allow a protein to denature during purification so columns are usually run at 4°C. Changes in pH, excessive dilution and addition of certain chemicals can also affect the stability of the protein.

Adsorption chromatography separates according to the affinity of the protein, or other material, for the surfaces of the solid matrix. Alumina (Al_2O_3), hydroxyapatite ($Ca_{10}(PO_4)_6(OH)_2$) or silica (SiO_2) are used for purifying non-polar molecules, whereas polystyrene based resins are effective matrices for polar molecules. This technique involves hydrogen bonding and/or van der Waals forces. The adsorbed proteins are usually eluted by increasing the ionic strength, often by using a gradient of increasing phosphate ion concentration.

Affinity chromatography is a powerful and highly selective purification technique. It can often give up to several thousand-fold purification in a single step. However, this chromatographic method is expensive on an industrial scale. The technique involves specific chemical interactions between solute molecules, such as proteins, and an immobilized ligand (functional molecule). Ligands are covalently linked to the matrix material e.g. agarose, via a spacer arm to avoid steric hindrance. Some ligands interact with a group of proteins, notably nicotinamide adenine dinucleotide, adenosine monophosphate, and Procion and Cibacron dyes; other ligands are highly specific especially substrates, substrate analogues and antibodies. Since monoclonal antibodies have become more readily available, immunoaffinity chromatography methods have been developed for the purification of various antigens.

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The loading capacity of affinity columns is large as the volume of the sample is unimportant and high resolution can be attained. This is also a high-speed technique and elution is achieved using specific cofactors or substrates alternatively non-specific elution may be performed with salt or pH change. For industrial scale operations, it is often necessary to sterilize the chromatographic media in order to comply with various regulatory requirements. This can be problematical as some ligands are sensitive to sterilizing agents.

Gel filtration chromatography essentially involves separation based on molecular size (molecular sieving), although molecular shape can also influence separation performance. Consequently, it is particularly useful for desalting protein preparations. The stationary phase consists of porous beads composed of acrylic polymers, agarose, cellulose, cross-linked dextran, etc., which have a defined pore size. These support materials should be sterilizable, chemically inert, stable, highly porous and hydrophilic, containing some ionic groups.

Mechanical rigidity is particularly important in order to maintain good flow rates. The initial choice of stationary phase material is also a key factor as some may interact with the target product, e.g. carbohydrate-based matrices interact with glycoproteins. Solute molecules below the exclusion size of the support material pass in and out of the beads. Molecules above the exclusion size pass only around the outside of the beads through the interstitial spaces and the apparent volume of the column is smaller for these larger excluded molecules. As a result, they flow faster down the column, separating from smaller molecules and eluting first. Smaller molecules able to enter the pores are then eluted in decreasing order of size.

Ion-exchange chromatography involves the selective adsorption of ions or electrically charged compounds onto ion-exchange resin particles by electrostatic forces. The matrix material is often based on cellulose substituted with various charged groups, either cations or anions. A commonly used example is the anion-exchange resin diethylaminoethyl (DEAE) cellulose. Proteins possess positive, negative or no charge depending on their isoelectric point (pI) and the pH of the surrounding buffer. If the pH of the buffer is below the pI, a protein has an overall positive charge, whereas a buffer at the pI results in the protein having no charge and pH above its pI produce an overall negative charge. For example, if a protein has a pI of 4.2, at pH 4.2 this protein is uncharged and will not bind to either positively or negatively charged resins. When the pH is raised to pH 7.0 the protein is negatively charged and binds to positively charged resins. In conditions below pH 4.0 the protein is positively charged and binds to negatively charged resins. If the protein is in anionic, state it is able to adsorb to DEAE cellulose. For example, any contaminants pass through the column product can then be desorbed as a purified fraction by altering the ionic strength of the buffer, often by using a gradient of increasing phosphate buffer concentration.

High-performance liquid chromatography (HPLC) was originally developed for the separation of organic molecules in non-aqueous solvents, but is now used for proteins in aqueous solution. This method uses densely packed columns containing very small rigid particles of 5-50μm diameter of silica or a cross-linked polymer. Consequently, high pressures are required. The method is fast and gives high resolution of solute molecules. Equipment for use in large-scale operations is now available.

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Hydrophobic chromatography relies on hydrophobic interaction between hydrophobic regions or domains of a solute protein and hydrophobic functional groups of the support particles. These supports are often agarose substituted with octyl or phenyl groups. Elution from the column is usually achieved by altering the ionic strength, changing the pH or increasing the concentration of chaotropic ions, e.g. thiocyanate. This technique provides good resolution and like ion exchange, chromatography has a very high capacity, as it is not limited by sample volume.

Metal chelate chromatography utilizes a matrix with attached metal ions, e.g. agarose containing calcium, copper or magnesium ions. The protein to be purified must have an affinity for this ion and binds to it by forming coordination complexes with groups such as the imidazole of histidine residues. Bound proteins are then eluted using solutions of free metal-binding ligands, e.g. amino acids.

MEMBRANE FILTRATION

Modern methods of filtration involve absolute filters rather than depth filters. These consist of supported membranes with specified pore sizes that can be divided into three main categories. They are, in decreasing order of pore size, **microfiltration**, **ultrafiltration** and **reverse osmosis** membranes. The suspension to be filtered is pumped across the membrane (cross/tangential flow) rather than at a right angle to it, it occurs in conventional filtration methods. This retards fouling of the membrane by particulate materials. Particles whose size is below the membrane 'cut-off' will pass through the membrane to become the ultrafiltrate or permeate, whereas the remainder is retained as the retentate. As filtration progresses, the flux across the membrane can slow due to membrane fouling. This may be caused by the accumulation of a layer of solute molecules on the surface of the membrane, referred to as concentration polarization. The presence of silicon antifoams may have a similar negative effect.

Microfiltration is used to separate particles of 10^{-2}µm to 10 µm including removal of microbial cells from the fermentation medium. This method is relatively expensive due to the high cost of membranes but it has several advantages compared with centrifugation. They include quiet operation, lower energy requirements, the product can be easily washed, good temperature control is possible, containment is readily achieved and no bioaerosols are produced. Consequently, it is suitable for handling pathogens and recombinant microorganisms.

Ultrafiltration is similar to microfiltration except that the membranes have smaller pore sizes, and are used to fractionate solutions according to molecular weight, normally within the range 2kDa to50kDa. The membranes have anisotropic structure, composed of a thin membrane with pores of specified diameter providing selectivity, lying on top of a thick highly porous support structure. A membrane manufactured with an exclusion size of 100000Da, should produce a retentate of proteins and other molecules over 100000Da and an ultrafiltrate of all molecules below 100000 Da. However, non-spherical proteins may exhibit different exclusion reactions to the membrane. Flat membranes are available for larger-scale operations hollow-fibre systems. Several of these ultrafiltration units can be linked together to produce a sophisticated purification

Career Avenues

system. These methods are extensively employed for the purification of proteins, and for separating and concentrating materials. Ultrafiltration is also effective in removing pyroxenes (bacterial cell wall lipopolysaccharides), cell debris and viruses from media, and for whey processing. Another variation on the ultrafiltration system is infiltration, where water or other liquid is filtered to remove unwanted low molecular weight contaminants. This can be used as an alternative to gel filtration or dialysis for removing ammonium sulphate from a protein preparation precipitated by this salt, for changing a buffer and in water purification.

Reverse osmosis is used for dewatering or concentration steps and has been employed to desalinate sea water for drinking. In osmosis, water will cross a semipermeable membrane if the concentration of osmotically active solutes, such as salt, is higher on the opposite side of the membrane. However, if pressure is applied to the 'salt side' then reverse osmosis will occur, and water will be driven across the membrane from the salt side. This reversal of osmosis requires a high pressure e.g. a pressure of 30–40 bar is needed to dewater a 0.6mmol/L salt solution (note: 1 bar=100kPa=0.987atm). Consequently, a strong metal casing is required to house this equipment. As the membranes have pore sizes of only 10.2 to 10.4 μm diameter, solute molecules can deposit on the surface, causing a large resistance to solvent flow. However, this fouling can be overcome by increasing the turbulence at the surface of the membrane.

Career Avenues

IMMOBILIZATION OF ENZYMES AND CELLS AND THEIR APPLICATION FOR BIOCONVERSION PROCESSES

Immobilization is the imprisonment of an enzyme in a distinct phase that allows exchange with, separated bulk phase in which the substrate, effector or inhibitor molecules are dispersed and monitored. Imprisonment refers to arresting the enzyme by certain means where polymer matrix is formed. The first commercial application of immobilized enzyme technology was realized in 1969 in Japan with the use *of Aspergillus oryzae* amino acylase for the industrial production of L-amino acids. Consequently, pilot plant processes were introduced for 6-amino penicillanic acid (6 APA) production from penicillin G and for glucose to fructose conversion by immobilized glucose isomerase.

ADVANTAGES OF USING IMMOBILIZED ENZYMES

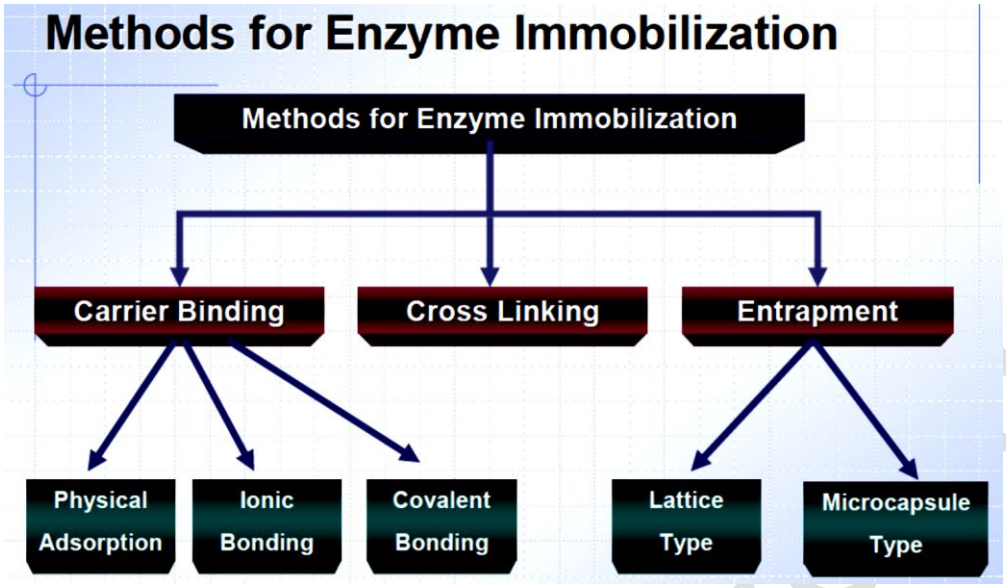
The advantages of using immobilized enzymes are:

(i) reuse (ii) continuous use (iii) less labor (iv) saving in capital cost (v) minimum reaction time (vi) less chance of contamination in products (vii) more stability (viii) improved process control and (ix) high enzyme to substrate ratio.

The first immobilized enzymes to be scaled up to pilot plant level and industrial manufacture are amino acid acylase, penicillin G-acylase and glucose isomerase. Some other industrially important enzymes are aspartase, esterase and nitrilase.

Career Avenues

METHODS OF ENZYME IMMOBILIZATION



CARRIER-BINDING

The carrier-binding method is the oldest immobilization technique for enzymes.

In this method, the amount of enzyme bound to the carrier and the activity after immobilization depend on the nature of the carrier.

The carrier-binding method can be further sub-classified into:

- Physical Adsorption
- Ionic Binding
- Covalent Binding

PHYSICAL ADSORPTION MODE

This method is based on the physical adsorption of enzyme protein on the surface of water-insoluble carriers.

ADVANTAGES

- Little or no conformational change of the enzyme or destruction of its active center
- Usually no reagents and only a minimum of activation steps are required
- Simple and cheap

Career Avenues

DISADVANTAGES

- Desorption of the protein resulting from changes in temperature, pH, and ionic strength
- Non-specific

IONIC BINDING MODE

This method relies on the ionic binding of the enzyme protein to water-insoluble carriers containing ion-exchange residues.

ADVANTAGES

- The conditions are much milder than those needed for the covalent binding method.
- Little changes in the conformation and the active site of the enzyme.
- High activities in most cases.

DISADVANTAGES

- Leakage of enzymes from the carrier may occur in substrate solution of high ionic strength or upon variation of pH

COVALENT BINDING MODE

The covalent binding method is based on the binding of enzymes and water-insoluble carriers by covalent bonds.

Covalent binding can be of following types:

- Diazotization : SUPPORT--N=N--ENZYME.
- Amide bond formation : SUPPORT--CO-NH—ENZYME
- Alkylation and Arylation: SUPPORT--CH_2-NH-ENZYME

  SUPPORT--CH_2-S—ENZYME
- Schiff's base formation : SUPPORT--CH=N—ENZYME
- Amidation reaction : SUPPORT--CNH-NH—ENZYME
- Thiol-Disulfide interchange : SUPPORT--S-S—ENZYME
- UGI reaction
- Mercury-Enzyme interchange
- Gamma-Irradiation induced coupling

Career Avenues

- Carrier binding with bifunctional reagents: SUPPORT-O $(CH_2)_2$ N=CH $(CH_2)_3$ CH=N-ENZYME

ADVANTAGES AND DISADVANTAGES OF COVALENT BINDING METHOD

ADVANTAGES

- Covalent binding is usually thought to be stable method by the enzyme carrier bond, which prevents elution of protein into the production stream.
- The wide range of choices is possible by selecting carrier materials and binding method. This allows a great deal of flexibility in designing an immobilized enzyme with specific physical and chemical properties, such as charge distribution, hydrophobe/hydrophile ration, spacer arm separation, partitioning capabilities, etc.

DISADVANTAGES

- Covalent methods are the relatively expensive and complicated in procedures.
- Activity yields may be low due to exposure of the enzyme to harsh environments or toxic reagent.

Active site may be modified through the chemical reactions used to create covalent bonding.

CROSS-LINKING

This method is based on the formation of covalent bonds between enzyme molecules by means of bi- or multifunctional reagent leading to 3-dimensional cross-linked aggregates.

ADVANTAGES

- Very little, desorption (as enzymes are strongly bound)
- Cross-linking is best used in conjunction with one of the other methods. It is used mostly as a means of stabilizing adsorbed enzymes and for preventing leakage.

DISADVANTAGES

- Cross-linking may cause significant changes in the active site of enzymes, and severe diffusion limitation may lead to significant loss of activity.
- Loss of enzyme activity during preparation.

The most common reagent used for cross-linking is Glutaraldehyde.

Career Avenues

Examples of immobilized enzymes in bioconversion:

A-CHYMOTRYPSIN ENCAPSULATION

A simple model reaction for the enzyme emulsion is the encapsulation of α-chymotrypsin, which catalyzes the hydrolysis of D, L-phenylalanine methyl ester into L-phenylalanine and the D-ester. The aqueous enzyme solution was immobilized in a suitable membrane phase. Stable emulsions can be obtained by kerosene or cyclohexane with 5 % Span 80 m the organic phase. The optimal working conditions are 25°C and pH 6 .The enzyme activity of the emulsion is about 30 % of the activity of the native enzyme.

L-LEUCINE DEHYDROGENASE ENCAPSULATTON

Reductive amination of α-ketolsocaproate by L-leucine dehydrogenase (LEUDH) produces L-Leucine during this reaction; while the coenzyme NADH, is oxidized and regenerated by a second enzyme formate dehydrogenase (FDH).

PENCILLIN-G-ACYLASE

The third enzyme emulsion type is encapsulating pencillin-G-acylase. In the transport mechanism, Pencillin G permeates through the membrane and the pencillin acylase splits the amide-bond to yield phenylacetic acid and 6-APA.

Career Avenues

AEROBIC AND ANAEROBIC BIOLOGICAL PROCESSES FOR STABILIZATION OF SOLID / LIQUID WASTES

BIOLOGICAL TREATMENT

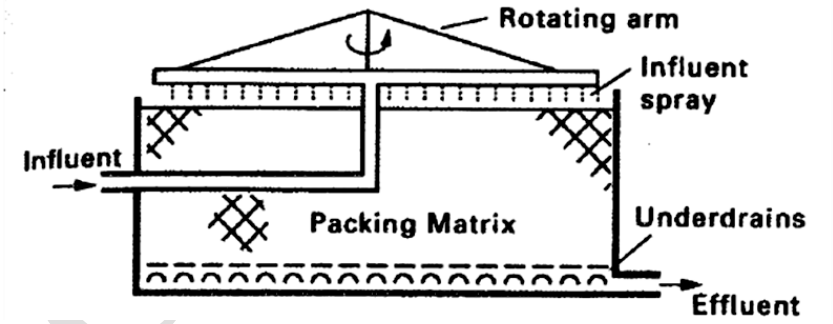
Most of organic-waste materials may be degraded biologically. This process may be achieved aerobically or anaerobically in a number of ways. The most widely used aerobic processes are trickling filters, rotating disc contactors, activated sludge processes and their modifications. The anaerobic processes include digestion, filtration and sludge blankets are used. Both the processes are adopted in the treatment of specific wastewaters and in sludge conditioning.

AEROBIC PROCESSES

TRICKLING FILTERS

The term filter in this unit operation is a misnomer, as the action of a trickling filter is not filtration, but rather it is a fixed film bio-reactor. Settled effluent to be treated is passed down through a packed bed counter-current to a flow of air. Microorganisms adhering to the packing matrix adsorb oxygen from the up flowing air and organic matter from the down flowing effluent; the latter is then metabolized and the effluent stream's BOD reduced.

A conventional trickling filter in the figure below usually consists of a cylindrical concrete tank 2 to 3 m in depth and 8 to 16 m in diameter.



Some filters are rectangular. Rotary system allows uniform hydraulic loading. The tank is packed with a bed of stone (usually granite) or special plastic pickings. The packing material diameter should be 50 to 100 mm to give a specific surface of around 100 m^2m^{-3}, and the material should be packed to give a voidage (percentage air space of total bed volume) of 45 to 55%, which should minimize the risk of the spaces between the packing materials becoming blocked by the microbial film. Synthetic packing material, although more expensive, has a higher surface area and voidage, allowing higher treatment rates per unit volume of bed and reducing the likelihood

of blockages. The trickling filter is always followed by a secondary sedimentation tank or humus tank to remove suspended matter from the treated effluent. In conventionally loaded or low-rate filters, the effluent from which the suspended solids have been removed is fed on to the upper surface of the bed by spray nozzles or mechanical distributor arms.

The effluent trickles gradually through the bed and a slime layer of biologically active material (bacteria, fungi, algae, protozoa and nematodes) forms on the surface of the support material. The large surface area created in the bed permits close contact between air flowing upwards through the bed. The bacteria in the biological film remove the majority of the organic loading. Complex organic materials are broken-down and utilized, nitrogenous matter and ammonia are oxidized to nitrates and sulphides. The higher organisms like protozoa control the accumulation of the biological film prevents the filter from blocking and improve the settling characteristics of the solids (humus) discharged with the filter effluent. In low-rate filters the scouring action of the hydraulic load normally has a minor role in removal of any loose microbial film. The active slime takes time to develop and can be poisoned by the addition of toxic chemicals. The simple filter is inefficient when operated at abnormally high organic loading rates. Initially, there is rapid buildup of bacteria, fungi and algae at the top of the filter, which cannot be controlled by the resident population of worms and larvae. The voids therefore block up resulting in ponding (untreated effluent accumulating on the surface of the filter bed). Film growth can be limited by reducing the dosing frequency, which gives better liquid distribution deeper into the bed.

A conventional trickling filter bed should remove 75 to 95% of the BOD and 90 to 95% of the suspended solids at organic loading rates of 0.06-0.12 kg BOD m^{-3} day^{-1}. When part of the treated effluent is being recirculated to dilute the feed and increase the hydraulic load placed on the unit, the organic loading can be increased to 0.9-0.15 kg BODm^{-3} day^{-1}. The increase in hydraulic load applied causes greater hydraulic scouring of the bed (preventing blockage), but does not reduce treatment efficiency due to improved wetting of the packing surface and thus, more efficient use of the biofilm.

To achieve the Royal Commission (20:30) Standard together with a high degree of nitrification, filters being supplied with domestic sewage should receive organic loading rates of 0.07-0.1 kg BOD m^{-3} day^{-1} and hydraulic loading rates of 0.12-0.6 m3 m^{-3} day^{-1}.

It is possible to modify the trickling filter to increase the capacity for organic loading by the use of two sets of filters and settling tanks in series; this is known as alternating double filtration (ADF). Effluent is applied to the first filter at a high hydraulic and organic loading rate, it passes from this filter through the first settling tank and then on to a second filter and settler. After a period of one to two weeks the sequence of the filters is reversed and the second filter receives the higher loading. In this way heavy film growth is promoted in the first filter to receive the effluent, but when the filter sequence is reversed it becomes nutrient limited, encouraging excess film removal. Loading rates of0.32-0.47 kg BOD m^{-3} day^{-1} have been claimed but recommended rates for design purposes are 0.15-0.26 kg BOD m^{-3} day^{-1}. Alternately, enclosed deep beds of 3.5 to 5.5-m depth may be used in which air is blown through the beds by fans.Cook (1978) has stressed the need to consider the possible intermittency of a factory wastewater treatment process. It was shown that starving of a laboratory-scale trickling filter beyond 48 hours resulted

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in near failure of the filter. This indicated the need to supplement or artificially load the filter to maintain available biomass in the system.

TOWERS

Because trickling filters do not have both a high specific area and a high voidage, they are less suitable for the treatment of large volumes of strong industrial effluents. Large areas of land would be required for the expensive and extensive traditional filter beds. Towers, 6-9 m in height, packed with lightweight (40-80 kg m^{-3}) plastic multi-faced modules or small random packing units have provided to be a space saving and relatively inexpensive solution to the problem. These pickings have a relatively open structure for oxygen transfer (specific surface of 100-300 m^2m^{-3}) and high voidage (90-98%), but are expensive compared with the conventional filter pickings. They are capable of coping with high BOD loadings. At a loading of 3.2 kg BOD m^{-3} day^{-1} a 50% BOD removal may be achieved and at 1.5 kg $m^{-3}day^{-l}$ 70% removal is possible. The biological film is similar to that formed on the conventional packing and scouring is due to the hydraulic load applied rather than the predation of higher organisms.

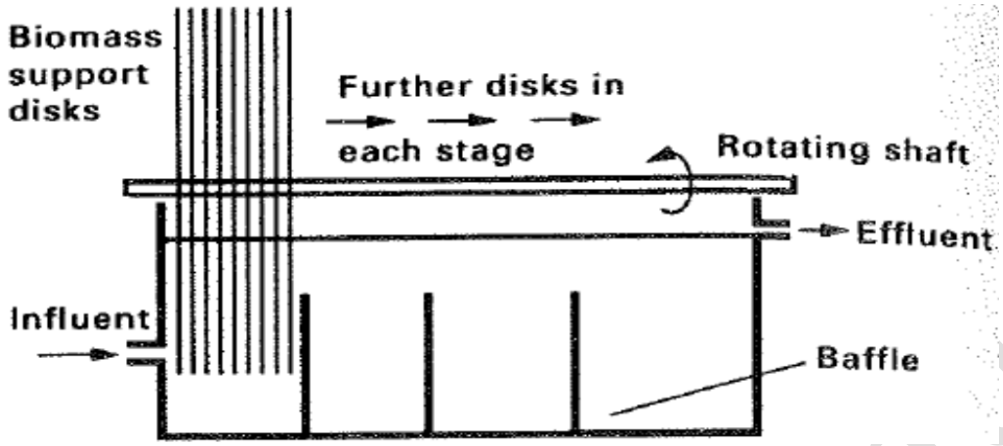
BIOLOGICALLY AERATED FILTERS (BAFS)

Biologically aerated filters are a relatively recent development based on the trickling filter. They consist of a packed bed which provides sites for microbial growth through which air is passed but, unlike trickling filters, the reactor volume is flooded with the effluent to be treated which is passed upwards or downwards through the reactor (i.e. co- or counter-currently to their supply) depending on the design. The packing matrix may be natural (e.g. pumice) or synthetic (e.g. polyethylene), and may be either a fixed structure or randomly packed.

The combination of aeration and filtration allows high rates of BOD and ammonia removal together with solids capture, so that sedimentation tanks may not be required. However, regular backwashing is essential to remove filtered solids and excess biomass. Organic loading rates for 90% BOD removal are significantly greater than those obtained for trickling filters, being in the range 0.7-2.8 kg BOD m^{-3} day^{-1}. They are versatile treatment systems, and are currently in operation, design capacities vary between 600 and 70,000 m^{-3} day^{-1}. In addition to their use as a secondary treatment process they can also be utilized for tertiary treatment or modified to allow de-nitrification in a manner similar to that of activated sludge systems.

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ROTATING BIOLOGICAL CONTACTORS (ROTATING DISC CONTACTORS)



In this treatment method a unit composed of closely spaced discs (2 to 3 m diameter with 1 to 2 cm spacing between discs), on a central drive shaft are rotated slowly (0.5 to 15 rpm) through the effluent so that 40 to 50% of the disc surfaces are submerged.

Figure - Rotating biological contracters

The discs, usually made from synthetic material (e.g. polystyrene, PVC), are arranged in stages or groups separated by baffles to minimize short circuiting or surging and to enhance specific treatment requirements such as nitrification. The discs may be flat or corrugated to increase surface area. A microbial film forms on the discs; this is aerated during the exposed part of the cycle and absorbs nutrients during the submerged part.

Shear forces produced as the discs rotate through the liquid control the thickness of the biofilm, with excess biofilm being sloughed from the discs. A sedimentation tank following the biological stage is therefore required to remove biological solids. Loading rates of 13 g BOD $m^{-2}day^{-1}$ for domestic sewage and partial treatment of loads of 400 g BOD $m^{-2}day^{-1}$ have been used. To achieve the 20:30 standard the loading rate should not exceed 6 g BOD $m^{-2}day^{-1}$.Rotating biological contactors are compact, easily covered for health and aesthetic reasons, available as packaged units, simpler to operate under varying loads than trickling filters (the biofilm being wetted at all times) and are easily added onto existing treatment processes. As such they can provide a cost effective method of on-site treatment. Ware and Pescod (1989) describe the use of full scale anaerobic/aerobic rotating biological contactors for treating brewery wastewaters. Greater than 85% COD removal was obtained in the aerobic stage, but difficulties were experienced in maintaining anaerobic populations.

Career Avenues

ROTATING DRUMS

Large rotating drums packed with random plastic packing materials or spheres have been manufactured as a development of rotating discs. Loading rates for the random packing were similar to those of rotating discs, while plastic spheres used in partial treatment could cope with loads of 6 kg BOD m^{-2} day^{-1}.

FLUIDIZED-BED SYSTEMS

Fluidized-bed reactors in wastewater treatment are relatively recent innovations. The support matrix (sand, anthracite, reticulated foam) has a large surface area on which the biofilm adheres and thus they are able to operate at high biomass concentrations with high rates of treatment. This allows strong wastewaters to be treated in small reactors. They are also useful for the treatment of industrial wastewaters when variable loadings are encountered (Cooper and Wheeldon, 1980, 1982). The support matrix is fluidized by the upflow of effluent through the reactor, and the degree of bed expansion is controlled by the flow rate of wastewater. The treated effluent can thus be decanted off without loss of the support matrix and with careful operation a secondary sedimentation tank may not be needed. The support matrix is regularly withdrawn to remove excess biomass. Fluidized-bed systems can be operated aerobically, anaerobically or anoxically for de-nitrification.

ACTIVATED SLUDGE PROCESS

The basic activated-sludge process consists of aerating and agitating the effluent in the presence of a flocculated suspension of micro-organisms on particulate organic matter.

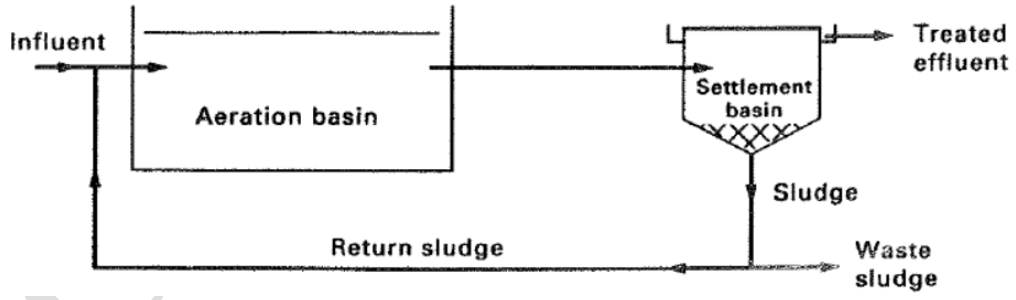


Figure: ACTIVATED SLUDGE PROCESSES

This process was first reported by Arden and Lockett (1914) and is now the most widely used biological treatment process for both domestic and industrial wastewaters. The raw effluent enters a primary sedimentation tank where coarse solids are removed. The partially clarified effluent passes to a second vessel, which can be of a variety of designs, into which air or oxygen is injected by bubble diffusers, paddles, stirrers, surface aerators etc. Vigorous agitation is used

to ensure that the effluent and oxygen are in contact with the activated sludge. After a predetermined residence time of several hours, the effluent passes to a second sedimentation tank to remove the flocculated solids. Part of the sludge from the settlement tank is recycled to the aeration tank to maintain the biological activity. The overflow obtained from the settlement tank should be of a 20:30 standard or better and be suitable for discharge to inland waters. The excess sludge is dewatered and dried, to be sold as a fertilizer, incinerated or land filled. In conventional activated sludge processes, organic loading rates are 0.5-1.5 kg BOD $m^{-3}day^{-l}$with hydraulic retention times of 5-14 hours depending on the nature of the wastewater, giving BOD reductions of 90-95%. High-rate activated-sludge processes can be used as a partial treatment for strong wastes prior to further treatment or discharge to a sewer and are widely used in the food processing and dairy industries.

The organic loading rate is 1.5-3.5 kg BOD $m^{-3}day^{-l}$, and with hydraulic retention times of only 1-2 hours, BOD reductions of 60-70% are possible (Gray, 1989).A number of modifications of the basic process can be used to improve treatment efficiency, or for a more specific purpose such as denitrification (Winkler and Thomas, 1978; Gray, 1989). Tapered aeration and stepped feed aeration are used to balance oxygen demand (which is greatest at the point of wastewater entry to the aeration basin) with the amount of oxygen supplied. Contact stabilization exploits biosorption processes and thereby allows considerable reduction in basin capacity (- 50%) for a given wastewater throughput. Denitrification (the biological reduction of nitrate to nitrite and on to nitrogen gas under anoxic conditions) can be accomplished in an activated-sludge plant when the first part of the basin is not aerated.

In advanced activated-sludge systems the amount of dissolved oxygen available for biological activity is increased to improve treatment rate. One vessel of this type is the 'Deep Shaft' (Hemming *et al.*, 1977), which is quite distinctive from the other aeration tanks and has been developed from the ICI PIC SCP process. The 'Deep Shaft' consists of a shaft 50 to 150 m deep, separated into a down-flow section (down-comer) and an up-flow section (riser). The shaft may be 0.5 to 10 m diameter, depending on capacity. Fresh effluent is fed in at the top of the 'Deep Shaft' and air is injected into the down-flow section at a sufficient depth to make the liquid circulate at 1 to 2 m $s^{-1.}$ The driving force for circulation is created by the difference in density (due to air bubble volume) between the riser and down-flow sections. For starting up, circulation of liquid is stimulated by injecting air at the same depth in the riser. Air injection is then gradually transferred to the air injection point in the down-comer. Because of the pressure created in the down-comer, oxygen transfer rates of 10 kg 2 $m^{-3}day^{-l}$ can be achieved and bubble contact times of 3 to 5 minutes are possible instead of 15 seconds in diffused air systems. BOD removal rates of 90% are achievable at organic loadings of 3.7-6.6 kg BOD $m^{-3}day^{-l}$ at hydraulic retention times of 1.17-1.75 hours (Gray, 1989). Sludge production was found to be much less than that for conventional sewage-treatment processes. Two types of pure oxygen systems have also been developed to increase the rate of oxygen transfer:

- Closed systems which operate in oxygen-rich atmospheres; and
- Open systems employing fine bubble diffusers.

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An example of the closed system is the UNOX Process developed by the Union Carbide Corporation in the U.S.A, and marketed in the U.K. by Wimpey Unox

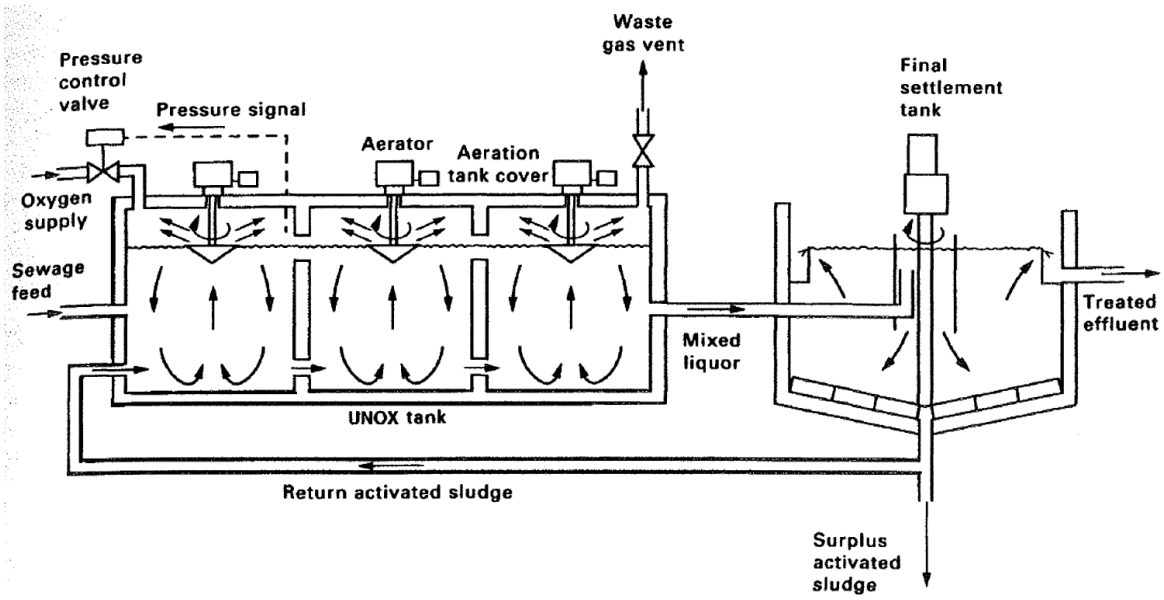
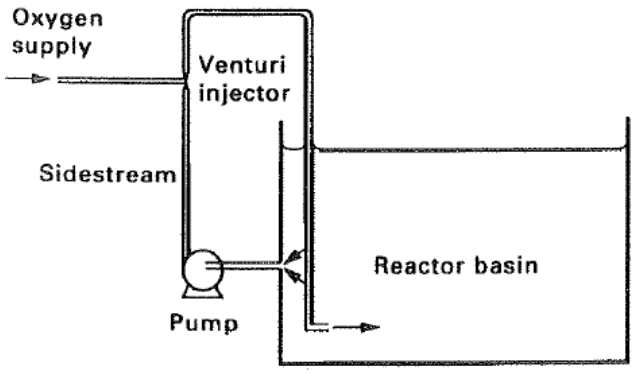


Figure: UNOX

The enclosed oxygenation tank is compartmentalized by baffles. Settled wastewater and returned sludge are fed into the first stage and oxygen pumped into the headspace. The oxygen and wastewater move sequentially through the compartments, and the oxygen concentration in the gas phase decreases in each stage as it is consumed by the micro-organisms. As the nutrient concentration also falls stage by stage, oxygen supply and demand is balanced. Organic loading rates (when treating municiple wastewater) are 3-4 times higher than those of aerated systems at 2.5-4.0 kg BOD $m^{-3}day^{-1}$.Pure oxygen systems also have shorter residence times and produce less sludge with better settling qualities than conventional systems. The UNOX process has been successfully used for the treatment of brewery wastewaters containing 2000 mg BOD dm^{-3} at flow rates of 2269 m^{-3} day^{-1} with a 6.6-hour retention time (Brooking et al., 1990).

The VITOX aeration system developed by the British Oxygen Company is an example of an open tank oxygenation system.



Career Avenues

Its main advantage is that it can be used in existing aeration tanks either to replace or upgrade conventional aeration without the need to replace or modify existing units; oxygenation is achieved by pumping settled wastewater at high pressure through a venturi where oxygen is injected. Turbulence and high pressures thus created ensure high levels of oxygen dissolution. The flexibility of this system is useful for the treatment of high strength intermittently produced wastewaters such as those generated by food processing (Gostick et al., 1990).

ANAEROBIC TREATMENT

Anaerobic treatment of waste organic materials originated with the use of septic tanks and Imhoff tanks, which have now been replaced by a variety of high-rate digesters (Pohland and Ghosh, 1971). Loehr (1968) has listed the following reasons for using anaerobic processes for waste treatment:

- Higher loading rates can be achieved than are possible for aerobic treatment techniques.
- Lower power requirements may be needed per unit of BOD treated.
- Useful end products such as digested sludge and/or combustible gases may be produced.
- Organic matter is metabolized to a stable form.
- There is an alteration of water-binding characteristics to permit rapid sludge dewatering.
- The reduced amount of microbial biomass leads to easier handling of sludge.
- Low levels of microbial growth will decrease the possible need for supplementary nutrients with nutritionally unbalanced wastes.

ANAEROBIC DIGESTION

Large volumes of wet sludge which are produced in primary and secondary sedimentation tanks may have to be reduced in volume before disposal. This volume of sludge can be reduced by anaerobic digestion. In sludges containing 20,000-60,000 mg dm^{-3} solid matter, 80% of the degradable matter may be digested, which will reduce the solids content by 50%. During anaerobic digestion acid fermenting bacteria degrade the waste to free volatile fatty acids, mainly acetic and propionic acid, which are then converted to methane (~ 60%) and carbon dioxide (~ 40%). The gas produced (biogas) is a very useful by-product, and can be burnt as a heating fuel, fed to gas engines to generate electricity or used as a vehicle fuel. As well as being used in sludge digestion and conditioning, anaerobic digesters are also used directly in the treatment of many high strength wastewaters, for example from the food and agricultural industries (Gray, 1989).

A number of anaerobic processes have been developed; completely mixed reactors (often these are simply described as anaerobic digesters) with or without sludge recycle, anaerobic filters, up-flow anaerobic sludge blankets (UASB) and anaerobic fluidized beds are the most common. Two

Career Avenues

stage anaerobic treatment systems utilizing a short retention time completely mixed acidification reactor followed by a UASB methanogenic reactor have been reported (Burgess and Morris, 1984; Ghosh et al., 1985). Benefits claimed include improved reliability, higher BOD,COD reductions and greater methane productivity.

ANAEROBIC DIGESTERS

The digester tanks used for this process may be up to 12,000 m^{-3} volume and equipped with heating coils for accurate temperature control normally in the mesophilic range (usually 28°-32° in anaerobic digestion) to increase the rate of digestion, which is also improved by mechanical agitation (Loll, 1977).

Some digesters are operated thermophilically (usually 55°-63° in anaerobic digestion) to increase the rate of degradation and biogas production, though much of the extra gas produced will be consumed in maintaining the digester temperature. The use of thermophilic anaerobic digestion has been investigated for the treatment of high strength (16-25 g COD dm^{-3}), low pH (≈3.8) and high temperature winery effluents (Romero *et al.,* 1988). Wiegant *et al.* (1985) reported the successful application of thermophilic anaerobic digestion of vinasse in completely mixed and UASB reactors, although methane production was found to decrease at high loading rates due to the presence of inhibitory compounds. Retention times are generally between 10 and 30 days, although with solids recycle retention times may be as low as 0.5 day. Most digesters operate on a semi-continuous basis with the appropriate volume of digested material being removed and replaced with 'fresh' material on a daily basis.

ANAEROBIC FILTERS

In anaerobic filters, as with the aerobic filters, there is a microbial film growing on an inert support (Chian and DeWalle, 1977). Anaerobic filters may be operated in an upflow or downflow mode, and a wide variety of pickings are used; both natural and synthetic (Young, 1983). The first full-scale anaerobic filters were constructed in the 1970s and in 1982 the Bacardi Corporation brought into operation a l00, 000 m^{-3}l to treat distillery effluents (Szendrey, 1983). Many other effluents are found to be amenable to treatment using anaerobic filter systems; antibiotic fermentation wastes, citric acid fermentation wastes, yeast production wastewater and brewery and winery wastewaters (Szendrey, 1983), molasses distillery slops (Silvero et at**.,** 1986) and fermentation and pharmaceutical wastes (Bonastre and Paris, 1989).

Career Avenues

BIOREMEDIATION

PCB BIODEGRADATION

Polychlorinate biphenols (PCB) are very toxic chemicals and suspected carcinogens. They include polychlorinated dibenzodioxins (PCDD), polychlorinated dibenzofurans (PCDF), and chlorinated benzenes. They have been found in sediments where they are a threat to the drinking water. A consortium of bacteria has been isolated that are able degrade PCBs. Some strains of bacteria were isolated that had varying degrees of competence in degrading PCBs. The bacteria metabolized the PCBs by using enzymes specific to the strain. In many cases a particular isomer may not be completely degraded, and a second species is needed to catalyze the degradation of the intermediate (although it cannot degrade the parent compound). Some 25 strains of indigenous and chemically mutated (grown naturally in diverse PCB sediments) have been isolated and tested. Of these *Alcaligenes eutrophs* and *Pseudomonas putida* degraded 13 different isomers to the extent of between 80 and 100%. Also *Arthrobacter* sps. were found to degrade most isomers. The degradation pathway may occur via oxidation of PCBs to chlorobenzoic acid. *Alcaligenes* are believed to have a dioxygenase that attacks the carbon at position 3,4. *Corynebacterium* sp MBl is believed to have a 2,3 dioxygenase. Effective strains usually contain plasmids. A mutant strain that has lost its plasmid can no longer metabolize PCBs. A few strains mineralize some chlorinated biphenols but in most cases a strain can degrade one ring of the PCB, but are unable to degrade the resulting chlorobenzoates which accumulate. *Pseudomonas* sp B13 is a hybrid that has the relatively non-specific oxygenase of *P.putida* and can completely mineralize the chlorobenzoates.

TCE DEGRADATION

Trichloroethene (TCE) is a common contaminant of our environment. It has been used extensively in the past in the dry cleaning industry. It is generally resistant to aerobic degradation. An anaerobic process will be described later. This anaerobic process has as its end product, vinyl chloride, a more potent carcinogen than TCE, and hence a different degradation pathway should be applied. Some methanotrophic bacteria which utilize methane have been found to degrade TCE. This is important since it is an aerobic process and one that might be able to be used in situ. Little *et al* isolated some strains of bacteria from a waste disposal site near Oak Ridge, TN. It had been used for direct dumping of chlorinated organic solvent wastes for a number of years. The cultures were grown and the pure methanotrophic strains isolated.

These were called 46-1 and 68-1. These appeared to be Type 1 methanotrophic bacteria based on their enzymatic activities and internal structure. Type 46-1 was chosen for the experiments due to its growth characteristics. In pure culture Type 46-1 converted 40% of TCE in a methane limited environment, incubated for twenty days. Most of the converted TCE was water soluble breakdown products, although some fraction was completely mineralized to carbon dioxide. It

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was found that the cells stopped growth and degrading TCE once the methane had been used. When the experiment was repeated but with the addition of more methane, more TCE was degraded.

The fact that methane or methanol is needed by the bacteria suggests a co-metabolic process. However, it seems that the methanotrophic bacteria alone cannot completely degrade TCE, but the heterotrophic bacteria normally associated with it finish the process. The apparent accumulation of glyoxylic acid, dichloroacetic acid, and carbon dioxide in TCE degrading cultures suggests a possible mechanism of biodegradation. The authors propose that the methane-oxidizing bacteria, such as Strain 46-1, convert TCE to its epoxide, which then breaks down spontaneously in water to form dichloroacetic acid, glyoxylic acid or one-carbon compounds such as formate or carbon monoxide. The two carbon acids accumulate in the water phase, while formate and carbon monoxide are further oxidized by methanotrophic bacteria to carbon dioxide. The hydrolysis of the TCE epoxide to form one carbon products explains the production of carbon dioxide from TCE by this methanotrophic organism. The degradation pathway can be seen in the figure 1.

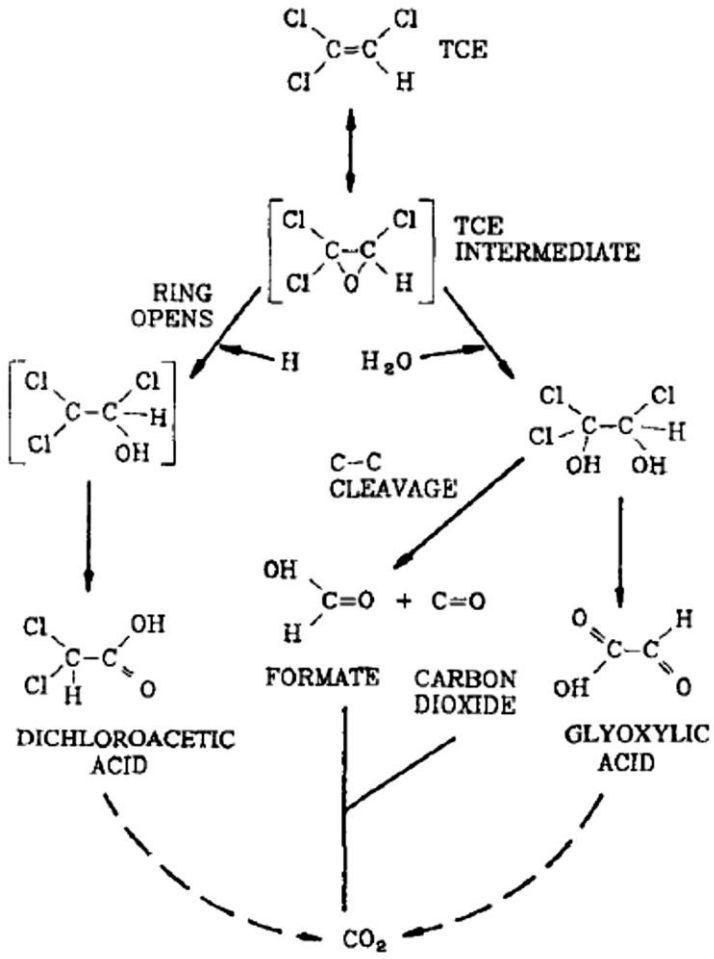


Figure 1: TCE Degradation by a Methane-Oxidizing Bacterium

POLYCYCLIC AROMATIC HYDROCARBON DEGRADATION

There have been numerous studies concerned with PAHs. These occur as natural constituents and combustion products of fossil fuels and are widespread environmental contaminants. They are toxic and have a potential for bioaccumulation and many are carcinogens. The most important mode of degradation may be via microorganisms. The ability to degrade PAHs depends on the complexity of their structure and the extent of enzymatic adaptation by bacteria. In general PAHs with 2 or 3 aromatic rings are readily degraded, but ones with 4 or more are usually recalcitrant and genotoxic. Meitkemp *et al.* isolated *Mycobacterium* sp, which readily degraded pyrene in pure culture under optimal conditions. This species was found in an ecosystem chronically exposed to petrogenic hydrocarbons. They treated both induced and non-induced cultures with chloramphenicol and showed that pyrene degrading enzymes were inducible in this *Mycobacterium* sp. This species could also degrade other PAHs and alkyl and nitro substituted PAHs such as naphthalene, phenanthrene, fluoranthenem 3-methylchloranethene, 1-nitropyrene, and 6-nitrochrysene.

ANTHRACENE DEGRADATION

Mahaffey *et al*. studied the degradation of benz(a)anthracene. The *Beirjerinkia* strain B1 used in this study was unable to utilize benz(a)anthracene as a source of carbon and energy for growth.

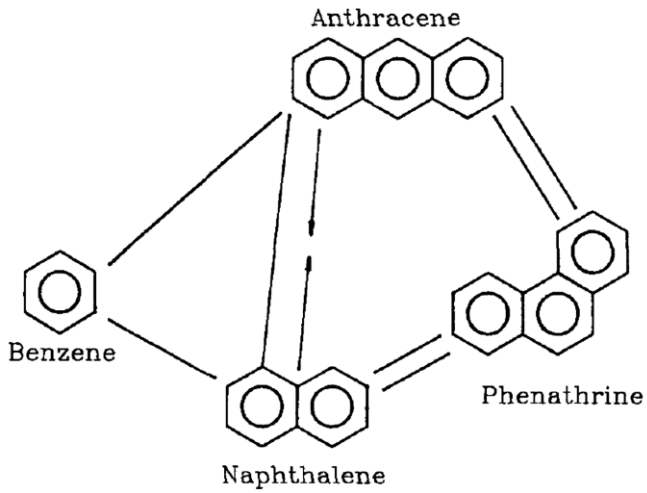
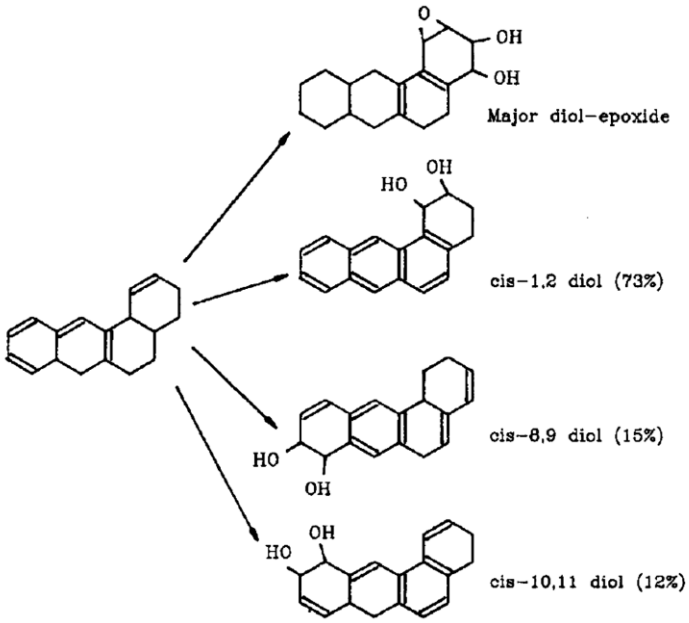


Figure 2. Proposed relationship between benzene and the PAHs used in the study.

Beirjerinkiu strain B8/36 converts most of benz(a)anthracene to the cis-1,2 diol is shown below.

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CHLOROPHENOL DEGRADATION

An interesting biodegradative technique, involving the aquatic macrophyte pennywort (Hydrocotyle umbrellutu) and bacteria was reported by Dieberg *et al*. In this study, they placed a thin layer of pennywort in a raceway and fed through sewage with 2,4 dichlorophenol (2,4-DCP) and monochloramine added to it. They found an acclimation to chlorophenol which suggests that the subsequent degradation was biotic. The plants developed masses of roots which together with the colonizing bacteria removed the toxins to extremely low concentrations; 100% for monochloramine and >95% for 2,4-DCP, even with high loading rates and low residence times over the sixty day study was observed. Pennywort only needs harvesting or thinning one or two times a year. Removal of contaminants such as 2,4-DCP and chloramines, can therefore be attained in pennywort thin film systems without incurring a major waste problem. The authors believe the phenol was primarily degraded to other compounds by the plant. They did not isolate the bacteria present, nor assess if the bacteria are necessary for the biodegradation; however, it is one of the rare studies involving plants as the degradative agents.

CHLORINATED WASTES

Highly chlorinated wastes are generally not considered readily biodegradable. Contaminated water may contain many of these carcinogens. Pelon and Mayo" wanted to see if bacteria would utilize various chlorinated wastes as the only carbon source. The wastes included, hexachlorobutadiene, 1,1,2-tricloroethane, tetra-chloroethane, hexachlorobenzene, and 1,2 dichloroethane. Bacteria from soil dump site were cultured with these organics with the only addition of salts. The experiments used aerobic mixing and a temperature of 25°C.

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They reported that the bacterial cultures were able to grow on the organic saturated groundwater with no added carbon source. They noted that the populations changed both qualitatively and quantitatively. The authors did not determine if indeed any of the compounds were mineralized, or if intermediate products were formed and they only hint at the presence of a consortium. Further information is needed, but the study does show that some bacteria can utilize these wastes.

P-NITROPHENOL DEGRADATION

Many factors limit the ability of bacteria to degrade compounds. If we are to treat hazardous wastes in a reliable manner, we need to know more about these factors. Yaldi *et al.* were interested in factors that limit the ability of bacteria added to lake water to stimulate biodegradation. ***Corynebacterium*** sp. were used for the study. They were made resistant to antibiotics by growing them in increasing concentrations of streptomycin, kanamycin, and streptomycin. The test compound was p-nitrophenol (PNP) which was added to water collected from lakes. The rate of biodegradation for low initial concentrations of PNP was found to be less than expected when extrapolated back from higher concentrations of PNP.

DEGRADATION OF FLUORO SUBSTITUTED BENZENES

Renganathen U. found that ***Pseudomonas*** sp strain T-12 cells in which the toluene degrading enzymes have been induced can transform many 3-fluoro-substituted benzenes to the corresponding 2,3-catechols with the concomitant release of inorganic fluoride. The substrates that induce 2,3-dioxygenase are 3-fluorotoluene, 3-fluorotrifluorotoluene, 3-fluorohalobenzene, 3-fluoronisole, and 3-fluorobenzonitrile. While 3-fluorotoluene and 3-fluoronisole produced only deflorinated catechols, other substrates led to catechol products both with and without the toluene substituent. The steric size of the C-1 substituent plays an important role in determining the deflourination reaction.

PENTACHLOROPHENOL DEGRADATION

Topp *et al*. investigated the interaction of pentachlorophenol (PCP) and an alternate carbon source on the survival and activities of Flavobacterium in axenic cultures. They found that PCB degradation does not require extra carbon source but the acclimation time is decreased if it is present.

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Oil Degradation

Emulsan is the extra cellular form of a heteropolysaccharides produced by the oil degrading bacterium *Acinetobacter calcoaceticus* RAG-1. It has potential use in the oil industry for example in reducing oil viscosity during pipe line transportation. Foght *et al*. studied the effect of emulsan on the biodegradation of oil by both pure cultures of bacteria and mixed cultures. They found a 50-90% decrease in degradation of alkanes after treatment with emulsan. The aromatic compound degradation was largely unaffected. The inhibition may be due to the emulsan stopping a direct physical interaction between the bacterial cells with the hydrophobic media that is necessary to initiate biodegradation. Several mechanisms are proposed; it could be due to a masking of hydrophobic sites on the bacterial cells. Further research is required if we are to successfully overcome this obstacle to biodegradation after oil is treated with emulsan. Although the emulsan makes transportation of oil easier, at present the choice of remediation by bacterial degradation in the case of a spill is weakend. Use of an emulsifying agent produced by a *Pseudomonas* strain has been demonstrated infield trials in the Middle East where when added to oil storage tanks for four days, it released 5600 barrels of oil that would normally have been lost and reduced the waste.

Aniline Degradation

In the environment not only one carbon source is usually available. This alternate substrate may either promote or inhibit the utilization of the chemical of concern. Konopka *et al*. was interested in the effect that a secondary carbon had on the degradation of aniline by a *Pseudomonas* sp that was isolated from sewage. Anilines are used in the manufacture of dyes, plastics, and pharmaceuticals. The strain K1 had a narrow range of aromatic substrate utilization. Resting cells from aniline grown cultures exhibited high respiratory activity upon the addition of aniline or catechol, some activity with toluidine, and no activity with the addition of a wide variety of other aromatic compounds, including dihydroxybenzylamine, chloroanilines, ethylanilines, aminophenols, aminobenzoates, and dihydroxybenzoates. The enzymes responsible for the degradation of aniline have some 2,3 dioxygenase activity.

Polyphosphate Degrading Enzymes

Acinetobacter sp is important in biological phosphate removal from wastewater.

Perchloroethylene

Methanosarcina mazei that is very common in methanogenic environments showed PCE degradation but worked best using methanol as the growth substrate.

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PAH DEGRADATION

Bumpus investigated the ability of *P.chrysosporim* to degrade a mixture of PAH since contamination is more likely to consist of more than one chemical and so if the fungus is to be of any practical use it must be able to degrade a mixture. The study used anthracene oil, a distillation product obtained by the fractional distillation of coal tar. At least 22 PAHs, including all the most abundant PAH components present in anthracene oil, underwent 70-100% disappearance during 27 days of incubation with nutrient limited cultures of this fungus. The lignin degrading enzyme system is expressed under nutrient limiting conditions. These results suggest that the fungus may be useful for the decontamination of sites contaminated with PAHs.

GATE 2010

Q.5 Match Group I with Group II

Group I	Group II
P. Staphylococcus aureus	1. Biofilms
Q. *Candida albicans*	2. Bacteriocins
R. *Mycobacterium tuberculosis*	3. Methicillin resistance
S.*Lactobacillus lactis*	4. Isoniazid

(A) P-1, Q-4, R-2, S-3
(B) P-2, Q-3, R-1, S-4
(C) P-3, Q-1, R-4, S-2
(D) P-1, Q-2, R-4, S-3

Answers:

- **Q.5 – B.**

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